

Polylogarithm Certificates and Sharp Euler Bounds

Fractional asymptotics and
polynomial Lerch zero thresholds

Complete proofs, exact certificates, and a revised research agenda

10 October 2026

Prepared for Vladimir Reshetnikov's ProveIt project

Source baseline. The manuscript and six pending incoming archives were inspected at commit `4c173c06cc32c9cea554b39be1837ad2ae897fc1`. The status claims in this article are relative to that snapshot.

Proof policy. Infinite statements are supported by ordinary mathematical proofs. Finite algebraic identities and sign enclosures have reproducible exact certificates. Numerical illustrations are identified separately. No numerical independence or proof-assistant formalization is claimed.

Abstract

This article continues the ProveIt polylogarithm manuscript and its six pending research archives at the explicitly pinned source revision of 10 October 2026. Four connected contributions are developed with complete proofs.

First, a finite rational tensor certificate proves Reshetnikov’s six-argument tetralogarithm conjecture. The same forty-two rows yield a one-parameter functional identity, an explicit further rational specialization, and a reusable descent procedure with several independent logarithms. Second, a compensated fractional-integral kernel gives the Euler remainder at every positive real order, including total orders below one. Its complete logarithmic expansion contains two distinct sequences of powers; off integral outer orders, the merged series at integral inner order diverges factorially.

The compensated kernel also determines the complete qualitative fixed-total comparison with the zero outer-order axis. Above and at outer order one there is strict domination at every truncation. Below one there is exactly one simple crossing in the continuous truncation-order interpolation. Exact rational arithmetic places the crossing for $a = b = 1/2$ strictly between 20 and 21.

Third, the conjectured sharp global late Euler asymptotic is proved: $C_N - 1 \sim cN^{-p}$, with the previously proposed explicit constants. In fact, for $N \geq \lceil e^{24} \rceil$, the supremum reduces exactly to the axis. A unique axis maximizer, its monotone motion, a fractional second asymptotic scale, and concentration of nearly maximizing parameters are established. Finally, classical multiplier and mesh preservation give an explicit reciprocal-gamma polynomial gap and a uniform Lerch zero threshold of order $n^6 \log^2 n$, improving the previous sufficient order $n^{2n+O(n)}$.

The accompanying files separate exact algebraic and interval certificates from numerical illustrations. The article records the resulting source status changes, a correction to an unsupported nonreducibility claim, and precise remaining questions. Novelty assertions are relative to the inspected project material; no general arithmetic independence or worldwide priority claim is made.

Keywords. Classical polylogarithms; tetralogarithm functional equations; Euler transformation; fractional integrals; Hurwitz zeta; reciprocal-gamma Appell polynomials; Lerch transcendent; zero separation.

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1 Scope, source baseline, and principal results

The problem is to turn specific numerical conjectures and qualitative asymptotic questions in the ProveIt project into exact mathematics. The source baseline is commit `4c173c06cc32c9cea554b39be1837ad2ae897fc1`, including the manuscript directory and all six archives then present in `docs/incoming` [1]. The supplied `data/source_manifest.json` records that commit, the archive hashes, and the principal source files used here. An *open question* below means open in that inspected material, unless a broader literature statement is explicitly supported.

The article concentrates on four targets with a direct connection to that material: a rational tetralogarithm formula, fractional Euler remainders and their extremal constants, comparison at fixed total order, and uniform saturation of Lerch zero counts. This focus permits full proofs and reproducible finite certificates. The higher Gaussian S_{2m} reductions and the higher golden ladders remain distinct problems.

1.1 From a numerical relation to an exact functional identity

The rational weight-four formula in `03-algebraic.tex`, label `golden:eq:tetra`, was proposed in 2015 and repeated in 2017 [5, 6]. The manuscript reduces it to a particular cross-base identity, but does not derive that identity with its required branch information. Theorem 2.1 completes the proof.

The certificate has forty-two rational functions of one variable. Their exact tensor cancellation implies constant identities for three modified real polylogarithms. A descent through weights four, three, and two then restores the ordinary tetralogarithms, with all endpoint and inversion contributions retained. Theorem 2.4 states the resulting functional family for every $0 < t < 1$. The original conjecture is its $t = 1/2$ consequence, while Corollary 2.5 gives a further explicit twenty-five-argument identity at $t = 1/3$.

The finite certificate is part of the proof: its complete coefficient table is printed, and its exact integer-coordinate cancellation is replayable. Numerical polylogarithm evaluations serve only to check the branch conventions independently.

1.2 Two parameter limits that require different arguments

Let $g_{a,b}$ be the imaginary part at i of the depth-two harmonic polylogarithm, let $E_N(a, b)$ be its N -term Euler transform, and write

$$R_N(a, b) = 2^N(E_N(a, b) - g_{a,b}), \quad C_N = \sup_{a,b>0} R_N(a, b).$$

Precise definitions are given in (3.1). For fixed positive a, b , the new leading law is

$$E_N(a, b) - g_{a,b} \sim \frac{\sqrt{\pi}}{2^{a+b}\Gamma(a+b)} 2^{-N} N^{-1/2} (\log N)^{a+b-1}.$$

Theorem 3.5 supplies every logarithmic order, uniformly on compact positive parameter sets. Its two coefficient sequences explain both termination at integral parameters and the factorial divergence in Proposition 3.7.

Taking a supremum over the whole positive quadrant is a different limit. The incoming uniform-bounds report proves $C_N \rightarrow 1$ and proposes the more precise rate [2]. We prove

$$C_N - 1 \sim cN^{-p}, \quad p = \frac{\log 2}{\log(3/2)}, \quad c = (1 - q^{-1})q^{-p}, \quad q = \frac{\log 3}{\log 2}.$$

There is no contradiction with the fixed-parameter law: the maximizing inner order grows as $\log N / \log(3/2)$, and the outer order tends to zero. These parameters leave every compact positive set.

Theorem 5.4 proves the exact reduction

$$C_N = \max_{b>0} R_N(0, b) \quad \text{for } N \geq \lceil e^{24} \rceil = 26,489,122,130.$$

That explicit threshold is sufficient and conservative. Theorem 6.3 gives a second, strictly smaller scale and the displacement of the maximizing inner order. The strict monotonicity of the axis maxima holds already at every $N \geq 2$; its transfer to C_N uses the eventual reduction.

1.3 A complete fixed-total comparison

At fixed $w = a + b$, the Gaussian value at $N = 1$ is strictly below the axis value. It is tempting to extend this ordering to every truncation. The exact answer depends on the outer order. For $a \geq 1$, Theorem 4.1 proves the strict comparison at every N , together with a positive Hausdorff-moment representation of the difference. For $0 < a < 1$, Theorem 4.4 proves exactly one simple reversal in the natural continuous interpolation of N . For $a = b = 1/2$, exact rational enclosures show

$$R_N(1/2, 1/2) < R_N(0, 1) \quad (1 \leq N \leq 20), \quad R_N(1/2, 1/2) > R_N(0, 1) \quad (N \geq 21).$$

The infinite classification follows from the sign-change theorem and two finite certified signs. It is not inferred from a plot.

1.4 A polynomial sufficient order for all Lerch zeros

The reciprocal-gamma Appell polynomials are defined by

$$\frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}.$$

Theorem 7.2 proves $\text{mesh}(Q_n) \geq 4/[n(n-1)(n-2)]$ for $n \geq 3$. Combined with a gamma concentration estimate that uses the distances to all roots, this yields the sufficient derivative order

$$\mathcal{K}_n = \left\lceil (16 + 6n(n-1)(n-2)H_{n-1})^2 \right\rceil \sim 36n^6 \log^2 n.$$

For every $k \geq \mathcal{K}_n$ and every $0 \leq \rho \leq 1$, the Lerch derivative family has exactly n positive zeros, all simple. If $x_{n,1} > \cdots > x_{n,n}$ are the roots of Q_n , their locations obey

$$\left| \log \frac{a_{n,j,k}^\rho}{k e^{-x_{n,j}}} \right| < \frac{8}{\sqrt{k}}.$$

This permits a simultaneous range $n \leq c k^{1/6} / (\log k)^{1/3}$ for any fixed $c < 1$. The classical mesh-preservation theorems are credited to their sources [7]; the new contribution is their explicit application and the improved sign transfer.

1.5 Conventions and the role of computation

All logarithms of positive real numbers are natural logarithms. For real polylogarithm arguments outside the unit interval, $\ell_n(x) = \text{Re Li}_n(x)$ denotes the real part of the principal continuation. Reciprocal gamma factors are interpreted as entire functions, including their zeros at nonpositive integers. In the Euler sections a is the outer order, b the inner order, and

N the truncation index. In the Lerch section n is the logarithmic index and k the derivative order. The letter ρ there ranges over the full closed interval $[0, 1]$.

The proof of each infinite assertion is given in the text. The exact programs verify finite tensor identities, symbolic normalizations, rational root brackets, and sign enclosures. The floating-point program evaluates positive integrals to illustrate the rate of convergence; it does not certify the global supremum or an asymptotic error bound. No proof-assistant formalization is claimed. The archive contains the complete source, the generated PDF, the replay programs, frozen outputs, a source manifest, and proposed integration notes.

2 An exact proof of the rational tetralogarithm conjecture

The manuscript's equation `golden:eq:tetra` reproduces the six-term relation proposed by Vladimir Reshetnikov in 2015 and again in 2017 [5, 6]. Its present proof discussion reduces it to a cross-base Kummer specialization but does not prove that specialization. We fill this local gap. The proof below is self-contained apart from elementary differential identities for the classical polylogarithms and a small finite rational certificate, supplied in full. It also proves a one-parameter functional identity from which the rational evaluation follows. No claim of worldwide priority for the underlying classical functional-equation mechanism is made.

Theorem 2.1 (The six rational arguments). *Put $A = \log 2$ and $B = \log 3$. Then*

$$\begin{aligned} & 720 \operatorname{Li}_4\left(\frac{1}{2}\right) - 2160 \operatorname{Li}_4\left(\frac{1}{3}\right) + 2160 \operatorname{Li}_4\left(\frac{2}{3}\right) + 270 \operatorname{Li}_4\left(\frac{1}{4}\right) \\ & \quad + 540 \operatorname{Li}_4\left(\frac{3}{4}\right) + 135 \operatorname{Li}_4\left(\frac{1}{9}\right) \\ & = 19\pi^4 + 30(\pi^2 - B^2)(10A^2 - 12AB + 3B^2) - 30A^2(19A^2 - 24AB + 8B^2). \end{aligned} \quad (2.1)$$

In particular, the intermediate relation `golden:eq:kummer` in the manuscript is also proved, by the two elementary duplications already recorded there.

2.1 A real generalized Rogers function and its exact descent

Write $\ell_n(x) = \operatorname{Re} \operatorname{Li}_n(x)$ on the principal branch. For $x \in \mathbb{R} \setminus \{0, 1\}$ set $L = \log |x|$, $M = \log |1 - x|$, so $\ell_1(x) = -M$. Define

$$\begin{aligned} R_2(x) &= \ell_2(x) - \frac{1}{2}L\ell_1(x), \\ R_3(x) &= \ell_3(x) - L\ell_2(x) + \frac{1}{3}L^2\ell_1(x), \\ R_4(x) &= \ell_4(x) - L\ell_3(x) + \frac{1}{2}L^2\ell_2(x) - \frac{1}{8}L^3\ell_1(x). \end{aligned}$$

These functions extend continuously by $R_n(0) = 0$ and $R_n(1) = \zeta(n)$. Direct telescoping differentiation gives

$$dR_n(x) = \frac{(-1)^n(n-1)}{n!} L^{n-2} (L dM - M dL), \quad n = 2, 3, 4. \quad (2.2)$$

At a crossing of $x = 1$, all logarithmically singular terms have positive powers of L , which tend to zero; this proves the stated continuity and allows identities obtained on regular subintervals to be joined.

Let $V = \mathbb{Q}(t)^\times \otimes_{\mathbb{Z}} \mathbb{Q}$, written additively, and put

$$\beta_n[f] = f^{\otimes(n-2)} \otimes (f \wedge (1 - f)).$$

The sign of a rational function is torsion and vanishes in V . Rational prime factors do not vanish and must be retained.

Lemma 2.2 (Descent with several independent logarithms). *Let $f_i(t)$ be nonconstant rational functions with no zero or pole on an interval I , and let $v_i \in \mathbb{Q}$ satisfy $\sum_i v_i \beta_4[f_i] = 0$. For every linear functional $\eta : V \rightarrow \mathbb{R}$ and each $n = 2, 3, 4$,*

$$\sum_i v_i \eta(f_i)^{4-n} R_n(f_i(t))$$

is constant on I , with the continuous interpretation at $f_i(t) = 1$.

Proof. Contract $4 - n$ of the first two tensor slots with η . The result is $\sum_i v_i \eta(f_i)^{4-n} \beta_n[f_i] = 0$. Evaluate the remaining ordinary tensor slots by $g \mapsto \log |g(t)|$ and the wedge by $g \wedge h \mapsto \log |g| d \log |h| - \log |h| d \log |g|$. Equation (2.2) gives a zero derivative on every regular subinterval. Continuity joins the constants across the finitely many crossings of $f_i = 1$. $\square$

The point of the lemma is that $\eta(f_i)$ need not be a rational multiple of a single logarithm. At the final specialization $t = 1/2$, the needed values are independent linear forms in $A = \log 2$ and $B = \log 3$. Thus the one-logarithm golden descent does not have to be applied outside its stated hypotheses.

2.2 The complete finite certificate

For the following forty-two rows define

$$f_i(t) = \varepsilon_i 2^{d_i} t^{a_i} (1 - t)^{b_i} (1 + t)^{c_i}. \quad (2.3)$$

All $a_i \geq 0$, and all factors except the displayed sign are positive for $0 < t < 1$. The coefficients below are primitive integers.

i	v_i	ε_i	d_i	a_i	b_i	c_i	i	v_i	ε_i	d_i	a_i	b_i	c_i
1	-1	+	-1	0	0	1	22	3	+	-1	1	0	-1
2	-11	+	0	0	0	1	23	-4	+	0	1	0	-1
3	-6	-	0	0	0	1	24	6	-	0	1	0	-1
4	-3	+	1	0	0	1	25	1	+	1	1	0	-1
5	-4	+	0	0	1	-1	26	-3	+	-1	1	0	0
6	4	-	0	0	1	-1	27	-3	-	-1	1	0	0
7	-1	+	-1	0	1	0	28	6	+	0	1	0	0
8	22	+	0	0	1	0	29	-6	-	0	1	0	0
9	21	-	0	0	1	0	30	3	+	1	1	0	0
10	-3	+	1	0	1	0	31	3	-	1	1	0	0
11	4	+	0	0	1	1	32	1	+	-1	1	0	1
12	-1	+	0	0	2	1	33	1	-	-1	1	1	0
13	-1	-	0	0	3	0	34	3	+	0	1	1	0
14	-1	+	-1	1	-2	0	35	3	-	0	1	1	0
15	-6	-	0	1	-2	0	36	-4	-	0	2	-1	-1
16	-6	-	0	1	-1	-1	37	-3	-	0	2	-1	0
17	3	-	-1	1	-1	0	38	-1	+	1	2	-1	0
18	-21	+	0	1	-1	0	39	-3	+	0	2	0	-1
19	-7	-	0	1	-1	0	40	-1	+	1	2	0	-1
20	1	-	1	1	-1	0	41	1	+	0	3	-3	0
21	-1	-	-1	1	0	-2	42	1	-	0	3	-1	-2

Lemma 2.3 (Exact tensor cancellation). *The displayed rational functions satisfy*

$$\sum_{i=1}^{42} v_i \beta_4[f_i] = 0. \quad (2.4)$$

Proof. The complete factor list needed for all f_i and $1 - f_i$, modulo sign, is

$$2, \quad t, \quad t-1, \quad t+1, \quad t-2, \quad t+2, \quad 2t-1, \quad 2t+1, \quad t^2-t-1, \quad t^2-t+1.$$

Use their exponent vectors as coordinates in V . For example, the constant 2 occurs in

$$1 - \frac{1+t}{2} = \frac{1-t}{2}, \quad 1 - \frac{t}{2(1-t)^2} = \frac{(2t-1)(t-2)}{2(1-t)^2}.$$

The first two slots of β_4 are symmetric. For each of the ten unordered pairs of coordinates of $2, t, 1-t, 1+t$, multiply the corresponding exponents and the two-by-two alternating minors of the vectors of $f_i, 1-f_i$, and sum with coefficient v_i . Every resulting coordinate is zero. There are 230 coordinates touched by the finite sum.

The accompanying `tetralogarithm_certificate.json` gives exactly the forty-two displayed rows. The self-contained verifier independently factors every rational function, checks the polynomial factorizations by multiplication, retains the prime-2 coordinate, and performs the indicated sums over rational integers. It does not search for a relation or use numerical polylogarithms in this check. This is a finite exact verification of (2.4). $\square$

2.3 The functional identity proved by the certificate

For $0 < t < 1$, put

$$A = \log 2, \quad X = \log t, \quad Y = \log(1-t), \quad Z = \log(1+t),$$

and define

$$P_4 = \frac{A^4}{4} - \frac{A^3(Y+Z)}{6} + \frac{A^2(Y^2+Z^2)}{4} + \frac{5A(Y^3+Z^3)}{6} - XY(3Y+Z)(Y+Z) \\ + \frac{17Y^4+8Y^3Z+9Y^2Z^2+2YZ^3+2Z^4}{6}, \quad (2.5)$$

$$P_2 = -6A^2 - 4A(Y+Z) + \frac{5}{2}Y^2 + 8YZ - 8Z^2. \quad (2.6)$$

Theorem 2.4 (A rational-function identity). *With the forty-two rows of the table,*

$$\sum_{i=1}^{42} v_i \operatorname{Re} \operatorname{Li}_4(f_i(t)) - 4 \operatorname{Li}_4\left(\frac{1}{2}\right) = P_4 + \zeta(2)P_2 - \frac{71}{4}\zeta(4), \quad 0 < t < 1. \quad (2.7)$$

The principal real parts and continuous values are used when an argument is greater than one or equals one.

Proof. Fix $t \in (0, 1)$ and write $L_i = \log |f_i(t)|$. Apply Lemma 2.2 with the functional sending a prime or irreducible factor $2, s, 1-s, 1+s$ respectively to A, X, Y, Z , and extend it linearly to the factor space, assigning arbitrary values to its other basis elements. Unique factorization shows that such a functional exists, and its value on every $f_i(s)$ is L_i . This definition remains valid when one of the additional factors of $1-f_i(t)$ vanishes; no logarithm of zero is used to define the functional. The table functions have limits

$$e_i = \lim_{s \rightarrow 0^+} f_i(s) = \begin{cases} 0, & a_i > 0, \\ \varepsilon_i 2^{d_i}, & a_i = 0. \end{cases}$$

Thus for $n = 2, 3, 4$,

$$\sum_i v_i L_i^{4-n} R_n(f_i(t)) = \sum_{a_i=0} v_i L_i^{4-n} R_n(e_i).$$

The elementary triangular identity

$$\ell_4(x) = R_4(x) + LR_3(x) + \frac{1}{2}L^2R_2(x) + \frac{1}{24}L^3\ell_1(x)$$

then supplies an exact ordinary-polylogarithm identity. More explicitly, put $M_i = \log |e_i| = d_i A$ for $a_i = 0$, and set $Q_i = -M_i^3/8 + L_i M_i^2/3 - L_i^2 M_i/4$. The result is

$$\begin{aligned} \sum_i v_i \ell_4(f_i(t)) &= \sum_{a_i=0} v_i \{ \ell_4(e_i) + (L_i - M_i) \ell_3(e_i) + \frac{1}{2}(L_i - M_i)^2 \ell_2(e_i) + Q_i \ell_1(e_i) \} \\ &\quad + \frac{1}{24} \sum_i v_i L_i^3 \ell_1(f_i(t)). \end{aligned} \quad (2.8)$$

When $e_i = 1$, $Q_i = 0$ and the corresponding last term in braces is zero; no divergent value is assigned to $\ell_1(1)$. Likewise $L_i^3 \ell_1(f_i(t))$ is assigned its limiting value zero at $f_i(t) = 1$.

Only $e_i \in \{1, -1, 1/2, 2\}$ occur. The required lower values are

e	$\ell_1(e)$	$\ell_2(e)$	$\ell_3(e)$
-1	-A	$-\zeta(2)/2$	$-3\zeta(3)/4$
1/2	A	$\zeta(2)/2 - A^2/2$	$7\zeta(3)/8 - \zeta(2)A/2 + A^3/6$
2	0	$3\zeta(2)/2$	$7\zeta(3)/8 + 3\zeta(2)A/2$

and $\ell_2(1) = \zeta(2)$, $\ell_3(1) = \zeta(3)$. For completeness, the lower reflection and Landen identities are

$$\begin{aligned} \text{Li}_2(x) + \text{Li}_2(1-x) &= \zeta(2) - \log x \log(1-x), \\ \text{Li}_2\left(-\frac{x}{1-x}\right) &= -\text{Li}_2(x) - \frac{1}{2} \log^2(1-x), \\ \text{Li}_3(x) + \text{Li}_3(1-x) + \text{Li}_3\left(-\frac{x}{1-x}\right) &= \zeta(3) + \zeta(2) \log(1-x) \\ &\quad - \frac{1}{2} \log x \log^2(1-x) + \frac{1}{6} \log^3(1-x), \end{aligned}$$

for $0 < x < 1$. Differentiating the first two identities gives zero differences, with constants fixed at $x \rightarrow 0^+$. Using them in the derivative of the third gives zero as well, and the same endpoint fixes its constant. Substitution $x = 1/2$ proves the displayed half values. The values at 2 follow by real inversion. Thus these endpoint reductions require no integer-relation recognition.

The endpoint weight-four sum is exactly

$$\sum_{a_i=0} v_i \ell_4(e_i) = 4 \text{Li}_4\left(\frac{1}{2}\right) + \frac{A^4}{4} - 6\zeta(2)A^2 - \frac{71}{4}\zeta(4).$$

Substitute $L_i = d_i A + a_i X + b_i Y + c_i Z$ and the factorizations of $1 - f_i$ in (2.8). All logarithms of the six additional polynomial factors cancel, as do the $\zeta(3)$ terms. The remaining polynomial is precisely $P_4 + \zeta(2)P_2 - 71\zeta(4)/4$, which proves the theorem. The verifier expands both the master formula and the displayed P_4, P_2 independently and checks their exact polynomial difference is zero. $\square$

2.4 The rational specialization, including all inversion terms

For $r > 1$, the real inversion formulas at weight four are

$$\begin{aligned} \ell_4(r) + \text{Li}_4(r^{-1}) &= -\frac{\log^4 r}{24} + \zeta(2) \log^2 r + 2\zeta(4), \\ \text{Li}_4(-r) + \text{Li}_4(-r^{-1}) &= -\frac{\log^4 r}{24} - \frac{1}{2} \zeta(2) \log^2 r - \frac{7}{4} \zeta(4). \end{aligned}$$

They follow by repeated differentiation from the real logarithm and dilogarithm inversion laws, fixing the constants at $r = 1$. Negative arguments inside the unit interval are removed by

$$\mathrm{Li}_4(-r) = \frac{1}{8} \mathrm{Li}_4(r^2) - \mathrm{Li}_4(r), \quad 0 \leq r \leq 1.$$

At $t = 1/2$ one has $X = Y = -A$, $Z = B - A$ and $f_i(1/2) = \varepsilon_i 2^{d_i - a_i - b_i - c_i} 3^{c_i}$. Applying the three displayed rules to the left side of (2.8), after subtracting its endpoint weight-four sum and multiplying by 180, gives exactly the six tetralogarithms in (2.1). The remaining polynomial from those inversions is

$$\begin{aligned} I = & 120A^4 - 510A^3B + 765A^2B^2 - 510AB^3 + 150B^4 \\ & + \zeta(2)(90A^2 + 2880AB - 1980B^2) - 1710\zeta(4). \end{aligned}$$

The non-weight-four part of the specialized descent master formula is

$$\begin{aligned} D = & -450A^4 + 210A^3B + 225A^2B^2 - 150AB^3 + 60B^4 \\ & + \zeta(2)(1890A^2 + 720AB - 1440B^2). \end{aligned}$$

Consequently the six-term sum equals $D - I$, namely

$$\begin{aligned} & -570A^4 + 720A^3B - 540A^2B^2 + 360AB^3 - 90B^4 \\ & + \zeta(2)(1800A^2 - 2160AB + 540B^2) + 1710\zeta(4). \end{aligned}$$

Using $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ gives exactly (2.1), proving Theorem 2.1.

2.5 A further proved rational specialization

The functional identity also produces explicit identities beyond the six-term conjecture. The specialization $t = 1/3$ is convenient because all logarithms reduce to $A = \log 2$ and $B = \log 3$. The following corollary is a new entry in this report, with no claim of priority among classical tetralogarithm identities.

Corollary 2.5 (Twenty-five positive rational arguments). *For the argument-coefficient pairs (r, c_r) in the table,*

$$\begin{aligned} \sum_r c_r \mathrm{Li}_4(r) = & \frac{109}{3} A^4 - 116A^3B + 54A^2B^2 + 36AB^3 - 21B^4 \\ & + \zeta(2)(316A^2 - 408AB + 132B^2) + 53\zeta(4). \end{aligned} \quad (2.9)$$

All arguments lie in $(0, 1)$, and the integer coefficient list is primitive: its greatest common divisor is one.

r	c_r	r	c_r
1/1024	1	9/64	-6
9/1024	-1	1/6	16
1/81	1	2/9	8
1/64	-4	1/4	-107
1/36	-6	8/27	8
1/32	-8	1/3	80
4/81	3	3/8	64
1/16	9	4/9	24
1/12	-24	1/2	-200
64/729	-1	16/27	-8
3/32	8	3/4	112
1/9	-14	8/9	32
1/8	64		

Proof. Set $t = 1/3$ in Theorem 2.4. The logarithms are $X = -B$, $Y = A - B$, and $Z = 2A - B$. Apply the real inversion formulas to arguments whose absolute value exceeds one, then the duplication formula $\text{Li}_4(-r) = \text{Li}_4(r^2)/8 - \text{Li}_4(r)$ to negative arguments in $[-1, 0]$. Multiplication by eight clears the resulting coefficient denominators. Collecting equal positive arguments gives exactly the table, and collecting all inversion terms gives the displayed polynomial. The same certificate verifier performs these reductions over $\mathbb{Q}$ and checks that the ordinary-polynomial residual is zero. As an independent audit only, a direct evaluation of the reduced identity at ninety working decimal digits has absolute residual below 8×10^{-90} . $\square$

The argument list and polynomial are supplied separately in `tetralogarithm_one_third.json`, and are also included in the shared certificate data. The replay checks the original conjecture, the functional identity, and this corollary in a single run.

2.6 Verification, provenance, and further questions

The exact verifier finishes with a zero tensor residual, a zero ordinary-polynomial residual, and the six specified rational arguments with coefficients (720, -2160, 2160, 270, 540, 135). Separately, it evaluates the functional identity at $t = 1/5, 37/100, 1/2, 4/5$ using 90 working decimal digits. The largest numerical discrepancy is approximately 7.9×10^{-90} . These calculations check the branch conventions; the proof is the finite tensor cancellation and the analytic descent.

The original questions are <https://math.stackexchange.com/questions/1373123/> and <https://mathoverflow.net/questions/261408/>. The former still presents a proposed Kummer route, while the latter's answer identifies the separate golden weight-five ladders. Those two facts should not be conflated: the certificate here settles the rational weight-four formula directly. The method is compatible with the higher-Bloch-condition approach to functional equations, but no unproved period conjecture or transcendence assertion is used.

Three useful next problems are to decompose the forty-two rows into a small set of named classical Kummer specializations; to specialize the proved functional identity at other rational or quadratic arguments and systematically reduce the resulting baskets; and to search for similarly complete certificates for the unresolved golden weights six and seven. The Gaussian S_6 conjecture remains a distinct problem and is not promoted by this result. No minimality or arithmetic independence of the six rational tetralogarithms is asserted.

3 Fractional Euler tails at all logarithmic orders

For positive real a, b , put $w = a + b$ and

$$H_m^{(b)} = \sum_{r=1}^m r^{-b}, \quad F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n \quad (|z| < 1), \quad g_{a,b} = \operatorname{Im} F_{a,b}(i).$$

The last value uses principal analytic continuation. Define

$$f_n = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad d_j = \sum_{n=0}^j (-1)^n \binom{j}{n} f_n, \quad E_N = \sum_{j=0}^{N-1} 2^{-j-1} d_j, \quad R_N(a, b) = 2^N (E_N - g_{a,b}). \quad (3.1)$$

The axis $a = 0, b > 0$ is included by the same continuation. The established positive two-measure formula proves convergence of E_N and $0 < R_N < 57/50$ throughout the positive quadrant; it remains valid when the ordinary boundary series fails to converge. We use that exact continuation theorem, not a rearrangement of a divergent series.

The manuscript proves, for integral orders, that R_N is $N^{-1/2}$ times a polynomial in $\frac{1}{2} \log N$, up to smaller power scales [1]. The incoming uniform-bounds report asks for its fractional continuation [2]. The answer contains two interlaced sequences of logarithmic powers. One terminates at positive integral total orders; the other terminates at positive integral outer orders. Their distinction is responsible for a change in the ordering of the errors.

3.1 A compensated kernel without a finite-measure restriction

Define, for $T > 0$,

$$h_b(T) = \frac{1}{\Gamma(b)} \int_T^\infty \frac{v^{b-1}}{e^v - 1} dv, \quad I^a h_b(T) = \frac{1}{\Gamma(a)} \int_0^T (T-v)^{a-1} h_b(v) dv. \quad (3.2)$$

The first function is locally integrable at zero for every $b > 0$; it and each of its derivatives decrease exponentially at infinity, up to fixed powers. At $b = 1$, it is $-\log(1 - e^{-T})$.

Lemma 3.1 (Exact moments of the positive seed). *For every $b > 0$ and integer $j \geq 0$,*

$$\int_0^\infty T^j h_b(T) dT = \frac{(b)_{j+1}}{j+1} \zeta(b+j+1). \quad (3.3)$$

Proof. Tonelli's theorem changes the left side to $((j+1)\Gamma(b))^{-1} \int_0^\infty v^{b+j}/(e^v - 1) dv$. Expanding the denominator as a positive geometric series and integrating each gamma integral gives the result. The exponent $b+j+1$ is greater than one, so every displayed zeta value is in its absolutely convergent range. $\square$

Theorem 3.2 (A compensated representation at every positive order). *Set*

$$V_{a,b}(T) = -\frac{T^{a+b}}{\Gamma(a+b+1)} + I^a h_b(T), \quad K_{a,b}(T) = V'_{a,b}(T), \quad W_N(x) = \frac{(1-x)^N}{1+x}. \quad (3.4)$$

Then, for $a, b > 0$ and integers $N \geq 1$,

$$R_N(a, b) = - \int_0^\infty K_{a,b}(T) e^{-T} W_N(e^{-2T}) dT. \quad (3.5)$$

The integral is absolutely convergent, including when $a+b < 1$. On the axis the same formula holds with

$$K_{0,b}(T) = -\frac{T^{b-1}}{\Gamma(b)(1-e^{-T})}. \quad (3.6)$$

Proof. Splitting the fractional integral at $T/2$ and substituting $v = T - u$ in its upper part shows that it is differentiable for $T > 0$. Near zero the elementary bound on the Bose denominator gives

$$V_{a,b}(T) = O((T^{a+b-1} + T^a + T^{a+b})(1 + |\log T|)),$$

and differentiation of the same split gives

$$K_{a,b}(T) = O((T^{a+b-2} + T^{a-1} + T^{a+b-1})(1 + |\log T|)).$$

In particular, $TV_{a,b}(T) \rightarrow 0$ and $T|K_{a,b}(T)|$ is locally integrable. Polynomial bounds suffice at infinity. These estimates can be made uniform on compact positive parameter sets; at $b = 1$ the logarithm bounds the removable difference quotient.

For real $n > 0$, Tonelli gives the Laplace transform

$$\int_0^\infty e^{-nT} h_b(T) dT = \frac{1}{n\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-nv})}{e^v - 1} dv = \frac{\zeta(b) - \zeta(b, n+1)}{n}.$$

At $b = 1$ the numerator means $\psi(n+1) + \gamma$. The integral establishes the continuation for $0 < b \leq 1$ directly; the two zeta poles cancel. The usual Hurwitz recurrence [8] therefore yields

$$n \int_0^\infty e^{-nT} V_{a,b}(T) dT = n^{-a} \{\zeta(b) - \zeta(b, n)\} =: c(n).$$

For positive integer n , this is $n^{-a} H_{n-1}^{(b)}$, and $c(1) = 0$. Integration by parts against a test function vanishing at zero now gives

$$c(n) = \int_0^\infty K_{a,b}(T)(e^{-nT} - e^{-T}) dT. \quad (3.7)$$

The boundary term vanishes by $TV(T) \rightarrow 0$. This compensation is essential when the unweighted density is not integrable.

For $j \geq 1$, insert (3.7) in the finite binomial sum to obtain

$$d_j = \int_0^\infty K_{a,b}(T) e^{-T} (1 - e^{-2T})^j dT.$$

The absolute sum over $j \geq 1$, with weights 2^{-j-1} , is bounded by an integral of a constant times $|K(T)|e^{-T}(1 - e^{-2T})/(1 + e^{-2T})$, which is integrable at both endpoints. Sum the tail and use the established analytic Euler limit to obtain (3.5). For $a = 0$, differentiate $h_b(T) - T^b/\Gamma(b+1)$ and repeat the argument. This gives (3.6) and the axis formula. $\square$

Remark 3.3 (Critical atoms). When $a + b = 1$, the primitive can have a nonzero limit at zero. The derivative in (3.4) is the ordinary derivative on $(0, \infty)$; a possible distributional endpoint atom is invisible to the tests used here, all of which vanish at zero. The theorem does not assert that the earlier finite signed-measure threshold has changed. It provides a compensated representation for the particular coefficients and remainders needed in this section.

3.2 The fractional density expansion

We interpret $1/\Gamma(z)$ as its entire continuation, so it is zero at every nonpositive integer.

Theorem 3.4 (Two sources of large- T behavior). *For every integer $M \geq 0$, uniformly on compact subsets of $(a, b) \in (0, \infty)^2$,*

$$K_{a,b}(T) = -\frac{T^{w-1}}{\Gamma(w)} + \sum_{k=1}^M \frac{(-1)^{k+1} (b)_k \zeta(b+k)}{k! \Gamma(a-k)} T^{a-k-1} + O(T^{a-M-2}) \quad (3.8)$$

as $T \rightarrow \infty$. If a is a positive integer, the sum terminates at $k = a - 1$ and its remaining error is exponentially small times a fixed power of T .

Proof. In $I^a h_b$, split at $T/2$. Differentiating the lower half is legitimate with its moving boundary. In the upper half write the integral as $\int_0^{T/2} u^{a-1} h_b(T-u) du / \Gamma(a)$. Both this half's derivative and the boundary terms are exponentially small times a fixed power of T , since $T-u \geq T/2$. Consequently

$$(I^a h_b)'(T) = \frac{1}{\Gamma(a-1)} \int_0^{T/2} (T-v)^{a-2} h_b(v) dv + O(e^{-T/2} T^C)$$

for some fixed C ; the formula includes $a = 1$ by the zero reciprocal gamma factor. Expand $(1-v/T)^{a-2}$ to M terms on $0 \leq v/T \leq 1/2$. Taylor's remainder is bounded uniformly by a constant times $(v/T)^M$. The moments in Lemma 3.1 give

$$\frac{(-1)^j}{j! \Gamma(a-j-1)} T^{a-j-2} \int_0^\infty v^j h_b(v) dv \quad (0 \leq j < M).$$

Extending the moments to infinity changes only the exponential error. The next moment bounds the remainder by $O(T^{a-M-2})$. Substitute $k = j+1$ and differentiate the first term of V .

For integral $a \geq 2$, the relevant power $(T-v)^{a-2}$ is a polynomial. For $a = 1$, the derivative is exactly $h_b(T)$. These observations establish the termination statement, including the exponential remainder. All estimates and reciprocal gamma coefficients are uniform across positive integral parameter values. $\square$

For integral a, b , (3.8) agrees exactly with the manuscript's polynomial $P_{a,b}$. The companion verifier checks this coefficient identity at all $1 \leq a, b \leq 8$ by rational symbolic arithmetic; the general equality also follows immediately from $(b)_k/k! = \binom{b+k-1}{b-1}$.

3.3 All logarithmic orders and their coefficient generator

Define the analytic functions near $z = 0$

$$\mathcal{M}(z) = \frac{1}{2} \Gamma\left(\frac{1-z}{2}\right) = \sum_{j \geq 0} \mu_j z^j, \quad \mu_j = \frac{(-1)^j \Gamma^{(j)}(1/2)}{2^{j+1} j!}, \quad (3.9)$$

$$\mathcal{D}_b(z) = \zeta(b) - \zeta(b, 1+z) = \sum_{k \geq 1} \frac{(-1)^{k+1} (b)_k \zeta(b+k)}{k!} z^k, \quad q_r(b) = [z^r] \mathcal{M}(z) \mathcal{D}_b(z). \quad (3.10)$$

Here $\mathcal{D}_1(z) = \psi(1+z) + \gamma$. In particular, no pole at $b = 1$ occurs in any coefficient. Writing $h = \gamma + 2 \log 2$, the gamma derivative identities give the useful exact generating formula

$$\mathcal{M}(z) = \frac{\sqrt{\pi}}{2} \exp \left\{ \frac{h}{2} z + \sum_{r \geq 2} \frac{(1-2^{-r}) \zeta(r)}{r} z^r \right\}. \quad (3.11)$$

Thus every μ_j is positive and can be generated by a finite Bell-polynomial calculation [9].

Theorem 3.5 (All logarithmic orders for real parameters). *Let $L = \frac{1}{2} \log N$. For integers $J \geq 1$ and $M \geq 0$,*

$$\begin{aligned} \sqrt{N} R_N(a, b) &= \sum_{j=0}^{J-1} \frac{\mu_j L^{w-1-j}}{\Gamma(w-j)} - \sum_{r=1}^M \frac{q_r(b) L^{a-r-1}}{\Gamma(a-r)} \\ &\quad + O\left(L^{w-1-J} + L^{a-M-2}\right). \end{aligned} \quad (3.12)$$

The estimate is uniform on every compact positive parameter set. The first and second sums can be truncated independently.

Proof. In (3.5), set $e^{-T} = t/\sqrt{N}$. The exact kernel becomes $(1 - t^2/N)^N/(1 + t^2/N)$ and $T = L - \log t$. The contribution from $T < L/2$ is smaller than every logarithmic power: use

$$W_N(e^{-2T}) \leq (1 - e^{-2T}) \exp(-(N-1)e^{-2T}) \leq 2T \exp(-(N-1)/\sqrt{N})$$

and local integrability of $T|K(T)|$. Polynomial growth controls the rest of this interval. The contribution of $0 < t < N^{-1/4}$ is also smaller than every inverse power of L , after the common $N^{-1/2}$ factor: logarithmic moments over that shrinking interval are bounded by $N^{-1/4}$ times a fixed power of L .

On $N^{-1/4} \leq t \leq N^{1/4}$, one has $T \geq L/2$ and $|\log t| \leq L/2$. The kernel can be replaced by e^{-t^2} with an error bounded in integral by N^{-1} times a fixed power of L . For example, its pointwise difference is at most $CN^{-1}e^{-t^2}(t^2 + t^4)$ when $t^2/N \leq 1/2$. Expand each $(L - \log t)^\nu$ by Taylor's theorem; the remainder is bounded by the next logarithmic moment on this central interval. The required moments are

$$\int_0^\infty e^{-t^2} (\log t)^j dt = 2^{-j-1} \Gamma^{(j)}(1/2).$$

Extending the central integrals to these full moments introduces only the negligible tails already bounded. Applied to the first power in Theorem 3.4, this gives the first sum and its error. For the k th secondary power, retain j with $k + j \leq M$. The gamma denominator is $\Gamma(a - k - j)$, so grouping by $r = k + j$ gives exactly $q_r(b)$ in (3.10). All remaining secondary terms are $O(L^{a-M-2})$. Compact uniformity follows from the density estimate, Taylor bounds on compact exponent sets, and the entire reciprocal gamma function. $\square$

Corollary 3.6 (The fractional leading law). *For all fixed $a, b > 0$, including $a + b < 1$,*

$$E_N - g_{a,b} \sim \frac{\sqrt{\pi}}{2^{a+b}\Gamma(a+b)} \frac{2^{-N}(\log N)^{a+b-1}}{\sqrt{N}}. \quad (3.13)$$

The first contributions, with their actual exponent order left visible, are

$$R_N(a, b) \sim \frac{\sqrt{\pi}}{2\sqrt{N}} \left\{ \frac{L^{w-1}}{\Gamma(w)} + \frac{h}{2} \frac{L^{w-2}}{\Gamma(w-1)} + \frac{h^2 + \pi^2/2}{8} \frac{L^{w-3}}{\Gamma(w-2)} - b\zeta(b+1) \frac{L^{a-2}}{\Gamma(a-1)} + \cdots \right\}. \quad (3.14)$$

For nonintegral b , the two sets of powers do not coincide. The secondary leading term is $b+1$ powers below the overall leading term. Consequently it lies between the first and second inverse-log corrections when $0 < b < 1$. Omitting it is not an all-orders extension of the integer formula.

3.4 Exactly what terminates, and what diverges

For a positive integer inner order m , the two sequences combine into a single coefficient generator:

$$\mathcal{G}_m(z) = \mathcal{M}(z)(1 - z^m \mathcal{D}_m(z)), \quad \beta_j^{(m)} = [z^j] \mathcal{G}_m(z). \quad (3.15)$$

The all-orders expansion is then

$$\sqrt{N} R_N(a, m) \sim \sum_{j \geq 0} \frac{\beta_j^{(m)}}{\Gamma(a + m - j)} L^{a+m-j-1}.$$

The symbol $\sim$ here specifies finite truncation asymptotics; it does not assert convergence of the infinite series.

Proposition 3.7 (Factorial divergence off integral outer orders). *For fixed positive integral m and positive nonintegral a ,*

$$\frac{\beta_j^{(m)}}{\Gamma(a+m-j)} \sim \frac{\sin(\pi a)}{2\pi\Gamma(m)} \Gamma(j-a) \quad (j \rightarrow \infty). \quad (3.16)$$

Thus the logarithmic series diverges factorially, with coefficients of one eventual sign. When a is a positive integer it terminates at $j = a + m - 1$.

Proof. The only apparent positive singularity of $\mathcal{G}_m$ inside $|z| < 2$ is at $z = 1$. It is removable because the Hurwitz recurrence gives $\mathcal{D}_m(1) = 1$ and $\mathcal{M}$ has a simple pole there. At $z = -1$, $\mathcal{D}_m(z) = -(1+z)^{-m}$ plus a holomorphic function, and $\mathcal{M}(-1) = 1/2$. Hence its highest principal part is $(-1)^m/(2(1+z)^m)$. Subtract the full principal part. The remainder is holomorphic on any disk of radius less than two, so Cauchy's coefficient bound proves

$$\beta_j^{(m)} \sim \frac{(-1)^{m+j}}{2\Gamma(m)} j^{m-1}.$$

Reflection gives

$$\frac{1}{\Gamma(a+m-j)} = \frac{(-1)^{m-j} \sin(\pi a)}{\pi} \Gamma(j+1-a-m).$$

The gamma-ratio asymptotic converts the product with j^{m-1} to $\Gamma(j-a)$, proving (3.16). For integral a , the reciprocal gamma factors instead vanish identically after the stated degree. $\square$

4 An exact phase diagram for fixed-total Euler comparisons

Retain the compensated density $K_{a,b}$, the scaled remainder $R_N(a,b)$, and $w = a + b$ from the preceding fractional expansion. Let

$$D_{a,b}(T) = K_{a,b}(T) - K_{0,w}(T), \quad q(T) = 1 - e^{-2T}.$$

The exact remainder identity gives, for integers $N \geq 1$,

$$R_N(a,b) - R_N(0,w) = - \int_0^\infty D_{a,b}(T) \frac{e^{-T} q(T)^N}{1 + e^{-2T}} dT. \quad (4.1)$$

The formula also defines a natural interpolation in real $N \geq 1$. All integrals below are ordinary absolutely convergent integrals after this vanishing test factor is included. No unweighted finite-measure assertion is made in the subcritical case.

4.1 All truncation orders on and above the outer-order threshold

Theorem 4.1 (Axis domination and complete monotonicity). *For every $a \geq 1$, $b > 0$, and integer $N \geq 1$,*

$$R_N(a,b) < R_N(0, a+b).$$

Writing $B_N = R_N(0, a+b) - R_N(a,b)$ and $\Delta B_N = B_{N+1} - B_N$, one has the stronger inequalities

$$(-1)^r \Delta^r B_N > 0 \quad (r \geq 0, N \geq 1).$$

In particular, B_N is strictly decreasing and $B_N B_{N+2} > B_{N+1}^2$ for every $N \geq 1$.

Proof. The density formulas give the exact identity

$$D_{a,b}(T) = (I^a h_b)'(T) + \frac{T^{w-1}}{\Gamma(w)(e^T - 1)}. \quad (4.2)$$

For $a > 1$ the first term is $I^{a-1}h_b(T) > 0$, while for $a = 1$ it is $h_b(T) > 0$. Thus $D_{a,b}$ is strictly positive on $(0, \infty)$. Substitution in (4.1) proves the comparison. Taking finite differences under the integral gives

$$(-1)^r \Delta^r B_N = \int_0^\infty D_{a,b}(T) \frac{e^{-T} q(T)^N (1 - q(T))^r}{1 + e^{-2T}} dT > 0.$$

The same formula is a Hausdorff moment representation, with a positive measure supported throughout $(0, 1)$ after the change of variable $x = q(T)$. Strict log-convexity follows from strict Cauchy–Schwarz for that measure. Here $a + b > 1$, so its unweighted mass is also finite; the statements only require the moments starting at $N = 1$. $\square$

Proposition 4.2 (The smaller scale at the threshold). *For fixed $b > 0$, with $L = \frac{1}{2} \log N$,*

$$R_N(1, b) - R_N(0, b + 1) \sim -\frac{L^b}{2\Gamma(b + 1)N}.$$

Thus the comparison at $a = 1$ occurs on a smaller power of N than the $N^{-1/2}$ scale of the generic fractional correction.

Proof. At $a = 1$, (4.2) becomes

$$D_{1,b}(T) = h_b(T) + \frac{T^b}{\Gamma(b + 1)(e^T - 1)} = \frac{e^{-T} T^b}{\Gamma(b + 1)} \left(1 + O(T^{-1})\right) \quad (T \rightarrow \infty).$$

The first term $h_b(T) = O(e^{-T} T^{b-1})$ follows directly by integrating its exponentially decaying density. Set $t = \sqrt{N} e^{-T}$ in (4.1). On the central range, the additional e^{-T} factor changes $N^{-1/2}$ to N^{-1} and changes the limiting Gaussian weight to te^{-t^2} . The same dominated tail estimates used for the fractional expansion apply, giving

$$R_N(0, b + 1) - R_N(1, b) = \frac{1}{N\Gamma(b + 1)} \int_{N^{-1/4}}^{N^{1/4}} te^{-t^2} (L - \log t)^b dt + O(N^{-1} L^{b-1}),$$

All powers here have positive arguments. Taylor expansion on this central range and extension of the Gaussian logarithmic moments to $(0, \infty)$ give the leading term $L^b \int_0^\infty te^{-t^2} dt = L^b/2$. $\square$

4.2 Exactly one reversal below the threshold

The fractional integral admits a useful alternative expression. Put

$$\phi_a(u) = \begin{cases} 1 - (1 - u)^{a-1}, & 0 < u < 1, \\ 1, & u > 1. \end{cases}$$

When $0 < a < 1$, this function is strictly negative below one and strictly positive above one.

Lemma 4.3 (A density with exactly one change of sign). *For $0 < a < 1$ and $b > 0$, the difference density $D_{a,b}$ has exactly one zero $T_* = T_*(a, b)$ on $(0, \infty)$. It is positive for $T < T_*$, negative for $T > T_*$, and $D'_{a,b}(T_*) < 0$.*

Proof. First derive the exact identity

$$\Gamma(a)\Gamma(b)T^{1-w}D_{a,b}(T) = \int_0^\infty \frac{\phi_a(u)u^{b-1}}{e^{Tu}-1} du + \frac{B(a,b)}{e^T-1}, \quad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(w)}. \quad (4.3)$$

Indeed, Tonelli's theorem in the definition of $I^a h_b$ gives

$$I^a h_b(T) = \frac{1}{\Gamma(a+1)\Gamma(b)} \int_0^\infty \frac{v^{b-1}}{e^v-1} (T^a - (T-v)_+^a) dv.$$

Differentiation yields

$$(I^a h_b)'(T) = \frac{1}{\Gamma(a)\Gamma(b)} \left\{ \int_0^T \frac{v^{b-1}[T^{a-1} - (T-v)^{a-1}]}{e^v-1} dv + T^{a-1} \int_T^\infty \frac{v^{b-1}}{e^v-1} dv \right\}.$$

The cancellation in the first integral must be retained if $b \leq 1$. Its integrand is $O(v^{b-1})$ at zero, and its moving-endpoint singularity has exponent $a-1 > -1$. One may justify differentiation first with a lower cutoff in v , split at a fixed neighborhood of $v = T$, and then pass to the limit; the displayed bounds give locally uniform convergence of the derivative. Scaling $v = Tu$ and using (4.2) proves (4.3).

Let $A(T)$ denote the right side of (4.3). It is the transform by $K(T, u) = (e^{Tu} - 1)^{-1}$ of the signed measure

$$d\nu(u) = \phi_a(u)u^{b-1} du + B(a,b)\delta_1(du).$$

This measure is negative on $(0, 1)$ and positive on $(1, \infty)$, with a positive atom at the dividing point. All its transforms and their T derivatives converge absolutely on compact positive T intervals. At zero in u , the density is $O(u^b)$; its singularity at $u = 1$ is integrable, and the kernel controls infinity.

For $T_2 > T_1 > 0$, the kernel ratio

$$r(u) = \frac{K(T_2, u)}{K(T_1, u)} = \frac{e^{T_1 u} - 1}{e^{T_2 u} - 1}$$

is strictly decreasing in u . For example, differentiate its logarithm and use strict increase of $x/(1 - e^{-x})$ on $(0, \infty)$. If $A(T_1) = 0$, subtraction of the constant $r(1)$ gives

$$A(T_2) = \int_0^\infty [r(u) - r(1)]K(T_1, u) d\nu(u) < 0.$$

Both open intervals give strictly negative contributions, while the atom contributes zero. Reversing this comparison shows $A(T) > 0$ for every $T < T_1$. Thus A can have at most one zero, and that zero has the stated sign orientation.

The endpoint signs can also be checked explicitly. As $T \downarrow 0$,

$$D_{a,b}(T) \sim \begin{cases} \frac{a}{(1-b)\Gamma(w)} T^{w-2}, & 0 < b < 1, \\ \frac{T^{a-1} \log(1/T)}{\Gamma(a)}, & b = 1, \\ \frac{\zeta(b)}{\Gamma(a)} T^{a-1}, & b > 1. \end{cases}$$

For the first line, the leading term of h_b is $T^{b-1}/[(1-b)\Gamma(b)]$; differentiating its fractional integral and adding the last term of (4.2) produces the positive coefficient $a/(1-b)$, including $w = 1$. The other two lines follow from $h_1(T) \sim \log(1/T)$ and $h_b(0) = \zeta(b)$ for $b > 1$. At infinity, the density expansion gives

$$D_{a,b}(T) \sim \frac{b\zeta(b+1)}{\Gamma(a-1)} T^{a-2} < 0.$$

Continuity therefore gives a zero, and the comparison proves uniqueness.

Finally, the logarithmic derivative

$$s_T(u) = \partial_T \log K(T, u) = -\frac{u}{1 - e^{-Tu}}$$

is strictly decreasing in u . At the zero,

$$A'(T_*) = \int [s_{T_*}(u) - s_{T_*}(1)] K(T_*, u) d\nu(u) < 0.$$

The positive prefactor in (4.3) proves $D'_{a,b}(T_*) < 0$. $\square$

Theorem 4.4 (A unique truncation-order reversal). *For every $0 < a < 1$ and $b > 0$ there is a unique real number $\nu(a, b) > 1$ such that the interpolation in (4.1) satisfies*

$$R_s(a, b) - R_s(0, a + b) \begin{cases} < 0, & 1 \leq s < \nu(a, b), \\ = 0, & s = \nu(a, b), \\ > 0, & s > \nu(a, b). \end{cases}$$

Its derivative with respect to s is strictly positive at the zero. Consequently the comparison for integral truncation orders reverses exactly once, allowing equality at one integer if $\nu(a, b)$ is integral.

Proof. Write $G(s)$ for minus the interpolated difference, namely the integral on the right of (4.1) without the minus sign. It converges with its s derivative for all $s \geq 1$. Near zero one may use $D_{a,b}(T) = O((T^{w-2} + T^{a-1})(1 + |\log T|))$. The test $q(T)^s = O(T^s)$ makes both powers integrable because $s + w > 1$ and $s + a > 0$ for $s \geq 1$. Additional logarithms from the s derivatives are harmless.

For $s_2 > s_1 \geq 1$, the ratio of the weights is $r(T) = q(T)^{s_2-s_1}$, strictly increasing in T . At any zero $G(s_1) = 0$, subtracting $r(T_*)$ and using Lemma 4.3 gives

$$G(s_2) = \int D_{a,b}(T) \frac{e^{-T} q(T)^{s_1}}{1 + e^{-2T}} [r(T) - r(T_*)] dT < 0.$$

The same comparison backwards gives $G(s) > 0$ before that zero. At a zero, replacing the bracket by $\log q(T) - \log q(T_*)$ proves $G'(s) < 0$.

It remains to establish existence. The manuscript's fixed-total Gaussian monotonicity theorem [1, subpick:thm:fixed-total], together with $E_1 = 0$, gives $G(1) = R_1(0, w) - R_1(a, b) > 0$. Its short comparison can be recalled explicitly. For independent $T \sim \text{Gamma}(w, 1)$ and $R \sim \text{Beta}(a, b)$, the positive double kernel gives

$$-g_{a,b} = \mathbb{E} \frac{X(X+Y)}{(1+X^2)(1+Y^2)}, \quad X = e^{-RT}, \quad Y = e^{-T}.$$

For fixed $T > 0$, the derivative with respect to X is $(2X + Y - YX^2)/((1+X^2)^2(1+Y^2)) > 0$. Thus the integrand decreases strictly in R , and its expectation is below the $R = 0$ axis value. This proves the required initial sign. For an independent elementary proof of the eventual negative sign, absorb the $s = 1$ weight into the signed integrable density. Its positive part is supported below T_* , where $q(T) \leq q(T_*)$. Choose a fixed closed interval strictly above T_* ; the negative part has strictly positive mass there and $q(T) \geq q_1 > q(T_*)$. Thus, for suitable $C, c > 0$,

$$G(s) \leq Cq(T_*)^{s-1} - cq_1^{s-1} < 0$$

for sufficiently large s . Continuity gives a zero, and the preceding strict comparisons prove the full statement. $\square$

For every fixed $a > 0$, $a \neq 1$, the leading total-order powers cancel exactly in the density difference. Its first nonzero term is $b\zeta(b+1)T^{a-2}/\Gamma(a-1)$, with relative error $O(T^{-1})$. The Gaussian transfer in Theorem 3.5 therefore gives

$$R_N(a, b) - R_N(0, a+b) \sim -\frac{b\zeta(b+1)\sqrt{\pi}}{2\Gamma(a-1)}N^{-1/2}L^{a-2}. \quad (4.4)$$

In particular, the eventual positive side for $0 < a < 1$ is

$$R_N(a, b) - R_N(0, a+b) \sim -\frac{b\zeta(b+1)\sqrt{\pi}}{2\Gamma(a-1)}N^{-1/2}L^{a-2} > 0 \quad (0 < a < 1).$$

The unique-crossing proof itself needs only the exact densities and their sign structure, so it is logically independent of the detailed all-order logarithmic expansion.

Corollary 4.5 (A unique maximum and the derivative zero sequence). *For $0 < a < 1$ and $b > 0$, let $\mathcal{D}(s) = R_s(a, b) - R_s(0, a+b)$ denote the interpolation above. For every $r \geq 0$, the derivative $\mathcal{D}^{(r)}$ has exactly one zero ν_r on $[1, \infty)$. These zeros are simple and satisfy*

$$1 < \nu_0 < \nu_1 < \nu_2 < \cdots, \quad \nu_r \longrightarrow \infty.$$

The sign of $\mathcal{D}^{(r)}$ before its zero is $(-1)^{r+1}$, and its sign afterwards is $(-1)^r$. In particular, $\mathcal{D}$ increases to a unique positive maximum at ν_1 and then decreases to zero.

Proof. Set $\ell(T) = -\log q(T) > 0$ and

$$H_r(s) = \int_0^\infty D_{a,b}(T) \frac{e^{-T} q(T)^s \ell(T)^r}{1 + e^{-2T}} dT.$$

Then $\mathcal{D}^{(r)} = (-1)^{r+1}H_r$ and $H'_r = -H_{r+1}$. Every H_r has the same single-change density signs as $D_{a,b}$. The preceding ratio argument shows that it has at most one zero, with strictly negative derivative there, and support separation shows that it is negative for all sufficiently large s .

Theorem 4.4 supplies the first zero ν_0 . At any zero of H_r , one has $H_{r+1}(\nu_r) = -H'_r(\nu_r) > 0$. Thus H_{r+1} has exactly one zero $\nu_{r+1} > \nu_r$. This proves induction and all sign assertions. To see that the zero sequence diverges, fix $s \geq 1$. The negative part of the integrand is supported where $\ell(T) \leq \ell(T_*)$. Choose a positive-mass closed interval below T_* on which $\ell(T) \geq \ell_1 > \ell(T_*)$. Its positive contribution dominates the entire negative part as $r \rightarrow \infty$. Hence $H_r(s) > 0$ for large r , so $s < \nu_r$. Finally $\mathcal{D}(s) \rightarrow 0$ by dominated convergence using the $s = 1$ absolute integrable weight. $\square$

A precise next asymptotic question. For fixed $b > 0$, the unique crossing order tends to infinity as $a \uparrow 1$: continuity at each fixed s and the strict comparison at $a = 1$ imply this directly. Balancing the generic fractional correction against the critical N^{-1} term suggests a boundary regime for $\nu(a, b)$, but proving that regime requires an expansion uniform in $1 - a$ and N . It is not asserted here.

4.3 An exact discrete transition at half orders

Corollary 4.6 (The first positive integer comparison is 21). *Let $\Delta_N = R_N(1/2, 1/2) - R_N(0, 1)$. Then*

$$\Delta_N < 0 \quad (1 \leq N \leq 20), \quad \Delta_N > 0 \quad (N \geq 21).$$

In the continuous interpolation, $20 < \nu(1/2, 1/2) < 21$.

N	Outward rounded enclosure for Δ_N
1	$(-0.324097778244, -0.324097778243)$
8	$(-0.015731327706, -0.015731327705)$
16	$(-0.002351554584, -0.002351554583)$
20	$(-0.000325283276, -0.000325283275)$
21	$(0.000023323498, 0.000023323499)$
32	$(0.001999520010, 0.001999520011)$
64	$(0.002856035510, 0.002856035511)$
128	$(0.002540045225, 0.002540045226)$
256	$(0.001952284331, 0.001952284332)$

Table 1: Exact rational enclosures for the fixed-total half-order comparison. The certificate files retain all rational endpoints and 24 outward rounded decimal places. Only the signs at 20 and 21 are needed for Corollary 4.6; the remaining rows illustrate the subsequent positive maximum and decay.

Proof. The exact rational interval calculation supplied with the article gives

$$\begin{aligned} -0.000325283275869852646927 &< \Delta_{20} < -0.000325283275869852646926, \\ 0.000023323498144449794846 &< \Delta_{21} < 0.000023323498144449794847. \end{aligned}$$

Theorem 4.4 then proves the whole integer classification. $\square$

Here are the details of the finite certificate. For each $N \in \{20, 21\}$, put $M = N + 100$ and compute

$$2^N \{E_N(1/2, 1/2) - E_M(1/2, 1/2) - E_N(0, 1) + E_M(0, 1)\}.$$

The inherited bound $0 < R_M < 57/50$ [4] makes its distance from Δ_N less than $(57/50)2^{N-M}$. This is one radius: the difference of two numbers in $(0, 57/50)$ has magnitude less than $57/50$. The direct finite Euler weights are

$$E_N = \sum_{n=0}^{N-1} \frac{(-1)^n}{2^N} \left(\sum_{j=n+1}^N \binom{N}{j} \right) f_n.$$

For every positive integer r , integer square roots produce rational dyadic bounds on $r^{-1/2}$:

$$k = \left\lfloor \sqrt{\lfloor 2^{2p}/r \rfloor} \right\rfloor, \quad \frac{k}{2^p} \leq r^{-1/2} < \frac{k+1}{2^p}, \quad p = N + 200.$$

All harmonic sums, products, Euler weights, and tail additions then use rational interval arithmetic. No floating-point operation contributes to acceptance of the signs.

The threshold replay imports the adjacent general fractional verifier. Its dependency and source hash are recorded explicitly; it is not a second arithmetic implementation. The global sign theorem and the finite enclosure computation provide different parts of the proof.

5 The optimal late Euler constant: eventual axis reduction

The incoming uniform-bounds report [2] proves $C_N \rightarrow 1$ and an explicit lower asymptotic $\liminf N^p(C_N - 1) \geq c$. It leaves the matching upper asymptotic open. We prove the stronger fact that, after an explicit (deliberately conservative) index, the exact global supremum is the maximum on the zero outer-order axis. This also identifies the extremizing inner order and a second asymptotic scale.

Write

$$W_N(x) = \frac{(1-x)^N}{1+x}, \quad F_N(y) = W_N(e^{-2y}), \quad J_N(s) = \mathbb{E}F_N(T_s), \quad D_N(s) = 1 - J_N(s),$$

where T_s is a rate-one gamma variable of shape $s > 0$ and $T_0 = 0$. Thus $J_N(0) = 0$. Define

$$R_N(a, b) = 2^N(E_N(a, b) - g_{a,b}), \quad C_N = \sup_{a,b>0} R_N(a, b), \quad A_N = \max_{b>0} R_N(0, b).$$

The exact representation already established in the manuscript is

$$R_N(a, b) = \mathbb{E} \frac{F_N(T_a + U_b) - F_N(T_a)}{1 - e^{-U_b}}, \quad (5.1)$$

with independent rate-one gamma variables. Its positive gamma-series form, valid also when $b \leq 1$, is

$$R_N(a, b) = \sum_{r \geq 1} r^{-b} \mathbb{E}\{F_N(T_a + U_b/r) - F_N(T_a)\}. \quad (5.2)$$

For $b > 1$ this gives the useful upper bound

$$R_N(a, b) \leq \zeta(b) J_N(a + b). \quad (5.3)$$

Indeed F_N is increasing, so every first expectation in (5.2) is at most $J_N(a + b)$, while the subtracted expectations are nonnegative. The known strict estimate $R_N(a, b) < 1$ for $a \geq 1$, $b > 0$ is the first case of the manuscript's Theorem `rw:thm:euler`. For clarity, it can also be recovered from the compensated density. At $a = 1$,

$$K_{1,b}(T) = -\frac{T^b}{\Gamma(b+1)} + h_b(T).$$

Each of the two positive terms has $e^{-T} dT$ integral one; for h_b , interchange its positive integrals and use $(1 - e^{-v})/(e^v - 1) = e^{-v}$. Their strict overlap makes the positive and negative Jordan masses strictly less than one. For $a > 1$, convolution with $T^{a-2}/\Gamma(a-1)$ in the T variable preserves zero total weighted mass and contracts the $L^1(e^{-T} dT)$ norm. Since $0 \leq W_N \leq 1$, the exact remainder integral is bounded above by the negative Jordan mass, and hence is strictly less than one. This gives the needed bound directly for every real $a \geq 1$.

5.1 A gamma-transform shape principle

We shall use the following elementary observation twice. Let $h(y) > 0$ be continuous, increase strictly to one maximum, and then decrease strictly, with distinct finite endpoint limits $h(0)$ and $h(\infty)$. Suppose each horizontal level between the smaller endpoint limit and the maximum crosses h either once or twice, as its shape prescribes. Consider

$$H(s) = \frac{r^s}{\Gamma(s)} \int_0^\infty h(y) y^{s-1} e^{-ry} dy, \quad s > 0.$$

For the bounded functions used below, every derivative in s exists. At a stationary point s_0 , set $\lambda = H(s_0)$ and $f = h - \lambda$. Since $H(s_0) = \lambda$ and $H'(s_0) = 0$,

$$\int f(y) y^{s_0-1} e^{-ry} dy = \int f(y) \log y y^{s_0-1} e^{-ry} dy = 0.$$

If f has just one sign change at y_1 , multiplication by $\log y - \log y_1$ produces an integrand of one strict sign, a contradiction. If f has signs $-$, $+$, $-$ and zeros $y_1 < y_2$, then $-(\log y - \log y_1)(\log y - \log y_2)$ has the same signs. Its integral against $f(y) y^{s_0-1} e^{-ry} dy$ is strictly positive. The two vanishing integrals therefore imply $H''(s_0) < 0$. Consequently every stationary point is a nondegenerate maximum. There is at most one, and the minimum of H on any compact interval is attained at an endpoint. Existence of one value above both endpoint limits implies a unique global maximum.

Lemma 5.1 (Quasiconcavity of a tilted defect). *For every integer $N \geq 2$, the function*

$$Q_N(s) = 2^s D_N(s), \quad s > 0,$$

has no interior minimum; in particular, its minimum on every compact interval is attained at an endpoint.

Proof. It is the gamma transform of shape s and rate two of

$$\psi_N(y) = e^y \{1 - W_N(e^{-2y})\}.$$

This bounded positive function has endpoint limits one and zero. We check its strict one-maximum shape. Put $x = e^{-2y}$ and

$$B_N(x) = \frac{(1-x)^{N-1} \{1 + 2(N+1)x + (2N-3)x^2\}}{(1+x)^2}.$$

Direct differentiation gives

$$\psi'_N(y) = e^y \{1 - B_N(x)\}.$$

The numerator of the logarithmic derivative of B_N after multiplication by the positive denominator $(1-x)(1+x)\{1 + 2(N+1)x + (2N-3)x^2\}$ is

$$\begin{aligned} P_N(x) &= N+1 - (2N^2 + N+5)x \\ &\quad - (4N^2 - 3N - 7)x^2 - (2N-3)(N-1)x^3. \end{aligned}$$

For $N \geq 2$ every nonconstant coefficient is negative, so P_N decreases strictly and has exactly one zero in $(0, 1)$. Since $B_N(0) = 1$, $B_N(1) = 0$, it first increases and then decreases, and crosses the level one exactly once in $(0, 1)$. It follows that ψ_N first increases and then decreases. The preceding gamma-transform principle, at rate two, proves the claim. $\square$

5.2 Excluding bounded and moderately growing inner orders

Lemma 5.2 (An explicit region below one). *Let N be an integer with $N \geq e^{24}$ and write $\ell = \log N$. Then*

$$R_N(a, b) < 1 \quad (0 \leq a \leq 1, \quad 0 < b \leq 2\ell).$$

Proof. First suppose $0 < b \leq 3$. Direct differentiation yields, for $N \geq 2$,

$$0 \leq F'_N(y) \leq 4Nx(1-x)^{N-1} \leq 4, \quad x = e^{-2y}.$$

Couple independent gamma variables so that $T_a \leq T_1$ and $U_b \leq U_3$, and put $V = T_1 + U_3$. Since $u/(1-e^{-u}) \leq 1+u$, the exact kernel is at most

$$(1+U_3) \sup_{0 \leq y \leq V} F'_N(y).$$

On $V \leq 10$, set $z = Nx \geq Ne^{-20} \geq e^4 > 50$. Then

$$F'_N(y) \leq 4ze^{-z/2} \leq 200e^{-25}.$$

On $V > 10$ use $F'_N \leq 4$. Now V has shape four, and the conditional mean of U_3 given V is $3V/4$. The elementary integer-shape gamma tails give

$$\begin{aligned} R_N(a, b) &\leq 800e^{-25} + 4\{\mathbb{P}(T_4 > 10) + 3\mathbb{P}(T_5 > 10)\} \\ &= 800e^{-25} + \frac{25928}{3}e^{-10} < 1. \end{aligned}$$

All constants in this estimate are explicit; for instance $e > 27/10$ suffices to verify the last inequality.

Next suppose $3 \leq b \leq 2\ell$. By (5.3) and the increase of J_N ,

$$R_N(a, b) \leq \zeta(b)\{1 - D_N(b+1)\}.$$

We prove $2^b D_N(b+1) > 4$ on this entire interval by checking its endpoints and using Lemma 5.1.

At $b = 3$, Chebyshev's inequality gives $\mathbb{P}(T_4 \leq 8) \geq 3/4$. Since $W_N(x) \leq e^{-Nx}$,

$$D_N(4) \geq \frac{3}{4}\{1 - e^{-Ne^{-16}}\} \geq \frac{3}{4}\{1 - e^{-e^8}\} > \frac{1}{2}.$$

At $b = 2\ell$, use

$$1 - W_N(x) \geq 1 - (1 - x)^N \geq \frac{Nx}{1 + Nx} \geq \frac{Nx}{2} \quad (0 \leq x \leq N^{-1}).$$

Thus, for $s = 2\ell + 1$ and a gamma variable Z_s of shape s and rate three,

$$D_N(s) \geq \frac{N}{2} 3^{-s} \mathbb{P}(Z_s \geq \ell/2).$$

Its mean is $s/3$ and its variance is $s/9$, so

$$\mathbb{P}(Z_s < \ell/2) \leq \frac{4(2\ell + 1)}{(\ell + 2)^2} \leq \frac{1}{2} \quad (\ell \geq 24).$$

Writing $L = \log(3/2) < 5/12$, we conclude that

$$2^{2\ell} D_N(2\ell + 1) \geq \frac{N^{1-2L}}{12} > \frac{e^4}{12} > 4.$$

The bound $L < 5/12$ follows from the trapezoidal bound for $\int_1^{3/2} dt/t$. Lemma 5.1, applied on $4 \leq s \leq 2\ell + 1$, proves $D_N(b+1) > 4 \cdot 2^{-b}$ throughout. Finally,

$$\zeta(b) - 1 \leq 2^{-b} \left(1 + \frac{2}{b-1}\right) \leq 2 \cdot 2^{-b}.$$

Hence $\zeta(b)(1 - D_N(b+1)) < 1$, as required. $\square$

5.3 Outer-order monotonicity in the remaining region

For a bounded Lipschitz function f , gamma addition gives the exact shape-generator formula

$$\frac{d}{da} \mathbb{E}f(T_a) = \int_0^\infty \frac{e^{-u}}{u} \mathbb{E}\{f(T_a + u) - f(T_a)\} du. \quad (5.4)$$

It includes the right derivative at $a = 0$. Indeed, divide the gamma-addition identity at $a + h$ by h and use $h\Gamma(h) \rightarrow 1$. A Lipschitz bound at zero and boundedness at infinity justify dominated convergence.

Lemma 5.3 (An explicit strictly decreasing outer-order range). *If $N \geq e^{24}$, $b \geq 2 \log N$, and $0 \leq a \leq 1$, then*

$$\partial_a R_N(a, b) < 0,$$

with the one-sided interpretation at $a = 0$.

Proof. Put $t_0 = \frac{1}{2} \log N$. In (5.4) for J_N , restrict to $T_a \leq 1$ and $u \in [t_0 + 1, t_0 + 2]$. Gamma addition gives $\mathbb{P}(T_a \leq 1) \geq 1 - e^{-1}$. For $N \geq 16$,

$$F_N(T_a + u) - F_N(T_a) \geq F_N(t_0 + 1) - F_N(1) > \frac{1}{2}.$$

To check the last inequality, use $1 - W_N(x) \leq (N + 1)x$, $W_N(x) \leq e^{-Nx}$, and $7 < e^2 < 8$. Consequently

$$J'_N(a) \geq \frac{(1 - e^{-1})e^{-2}}{\sqrt{N}(\log N + 4)} > \frac{1}{18\sqrt{N}(\log N + 4)}. \quad (5.5)$$

For $b > 1$, rewrite the positive series as

$$R_N(a, b) = \zeta(b)\{1 - J_N(a)\} - \mathcal{D}_N(a, b), \quad \mathcal{D}_N(a, b) = \sum_{r \geq 1} r^{-b} \mathbb{E}\{1 - F_N(T_a + U_b/r)\}.$$

The derivative of $1 - W_N(x)$ lies in $[0, N + 1]$, and therefore

$$0 \leq F_N(y + u) - F_N(y) \leq (N + 1)e^{-2y}(1 - e^{-2u}).$$

Apply (5.4) to each summand. Positivity and the following summable bound justify interchanging derivative and sum:

$$0 \leq -\partial_a \mathcal{D}_N(a, b) \leq (N + 1) \log 3 \sum_{k \geq 3} k^{-b}. \quad (5.6)$$

Here $\int_0^\infty (e^{-u} - e^{-3u}) du/u = \log 3$, and $r^{-b} \mathbb{E}e^{-2U_b/r} = (r + 2)^{-b}$. Since $b \geq 2 \log N \geq 3$,

$$\sum_{k \geq 3} k^{-b} \leq 3^{-b} \left(1 + \frac{3}{b-1}\right) \leq 3 \cdot 3^{-b} \leq 3N^{-2},$$

where $3 > e$ was used in the last step. Thus the right side of (5.6) is at most $12/N$. By (5.5),

$$\partial_a R_N(a, b) < -\frac{1}{18\sqrt{N}(\log N + 4)} + \frac{12}{N} < 0.$$

The final inequality follows from $\sqrt{N} > 216(\log N + 4)$ for $\log N \geq 24$. $\square$

Theorem 5.4 (Eventual exact reduction of the global supremum). *For every integer $N \geq \lceil e^{24} \rceil$,*

$$C_N = A_N = \max_{b > 0} R_N(0, b).$$

Every positive outer order gives a strict inequality $R_N(a, b) < A_N$. If the axis $a = 0$ is admitted, the unique maximizing pair is $(0, b_N)$, where b_N is the unique maximum of the axis function proved below. The threshold is an explicit sufficient condition, not an estimate of the smallest index for which the conclusion holds.

Proof. The axis maximum is greater than one. This follows already from the exact Taylor lower bound

$$R_N(0, b) \geq 1 + 2^{-b} - N3^{-b} - N4^{-b} > 1$$

for all sufficiently large b at fixed N . For $a \geq 1$ the previously proved strict bound is $R_N(a, b) < 1$. For $0 \leq a \leq 1$ and $b \leq 2 \log N$ use Lemma 5.2. In the remaining range, Lemma 5.3 gives $R_N(a, b) < R_N(0, b) \leq A_N$ when $a > 0$. Finally the bounded kernel in (5.1) is continuous under $a \downarrow 0$, so the axis values are approached by positive outer orders. This proves equality of the suprema and all strict claims. Uniqueness on the axis follows from the next theorem. $\square$

6 The maximizing inner order and a fractional second scale

On the axis the exact integral is

$$A_N(b) := R_N(0, b) = \frac{1}{\Gamma(b)} \int_0^\infty y^{b-1} e^{-y} K_N(e^{-y}) dy, \quad K_N(u) = \frac{(1+u)(1-u^2)^{N-1}}{1+u^2}.$$

In this section $A_N(b)$ denotes the function and A_N without an argument its maximum.

Theorem 6.1 (A unique axis maximum at every index). *For each integer $N \geq 2$, $A_N(b)$ has exactly one stationary point $b_N > 0$. It is a nondegenerate global maximum, greater than one; A'_N is positive below b_N and negative above it.*

Proof. The sign of $K'_N(u)$ on $0 < u < 1$ is the sign of

$$1 - (2N+1)u + u^2 - (2N-3)u^3.$$

This polynomial decreases strictly, from a positive value to a negative value, because its derivative is at most $-(2N+1) + 2 < 0$. Thus $K_N(e^{-y})$ increases strictly from zero to one maximum, then decreases strictly to one. The gamma-transform principle proves that all stationary points of A_N are nondegenerate maxima. Moreover $A_N(b) \rightarrow 0$ as $b \downarrow 0$, $A_N(b) \rightarrow 1$ as $b \rightarrow \infty$, and the preceding Taylor lower bound gives values greater than one. A global maximum therefore exists, and the shape principle makes it unique. $\square$

Corollary 6.2 (Monotone height and maximizing inner order). *For every integer $N \geq 2$,*

$$A_{N+1} < A_N, \quad b_{N+1} > b_N.$$

In particular C_N is strictly decreasing for $N \geq \lceil e^{24} \rceil$.

Proof. The pointwise relation $K_{N+1}(u) = (1-u^2)K_N(u) < K_N(u)$ implies $A_N(b_{N+1}) > A_{N+1}(b_{N+1}) = A_{N+1}$, and hence $A_N > A_{N+1}$. For the locations, the ratio $A_{N+1}(b)/A_N(b)$ is the expectation of $1 - e^{-2T}$ under the probability density proportional to $K_N(e^{-T})T^{b-1}e^{-T}$. Its derivative in b is the strictly positive covariance of $1 - e^{-2T}$ and $\log T$ under that density. Thus $A'_{N+1}(b_N) > 0$, since $A'_N(b_N) = 0$. The unique-maximum theorem gives $b_{N+1} > b_N$. The eventual reduction transfers the height inequality to C_N . $\square$

Set

$$\begin{aligned} L &= \log(3/2), & B &= L^{-1}, & p &= B \log 2, & q &= \frac{\log 3}{\log 2}, & d_0 &= B \log q, \\ c &= (1 - q^{-1})q^{-p}, & \nu &= B - \frac{1}{2}, & I &= \frac{1}{2} - B + B \log(2B), \\ D &= \sqrt{\frac{B}{2\pi}} \Gamma\left(\frac{1}{2} - B\right) (2B)^{-d_0}, & K &= \frac{\log(2B)D}{(\log 2)Lq^{-p}}. \end{aligned}$$

The elementary bounds $2/5 < L < 5/12$ show $1 < \nu < 2$, so $D > 0$. Also $p < I < 2p - 1$.

Theorem 6.3 (The sharp global late asymptotic and maximizing order). *As $N \rightarrow \infty$, with $\ell = \log N$,*

$$C_N = A_N = 1 + cN^{-p} + D \frac{N^{-I}}{\sqrt{\ell}} + o\left(\frac{N^{-I}}{\sqrt{\ell}}\right), \quad (6.1)$$

$$b_N = \frac{\log(Nq)}{\log(3/2)} - K \frac{N^{-(I-p)}}{\sqrt{\ell}} + o\left(\frac{N^{-(I-p)}}{\sqrt{\ell}}\right). \quad (6.2)$$

In particular the incoming conjecture is settled:

$$\lim_{N \rightarrow \infty} N^p (C_N - 1) = c.$$

The numerical diagnostics are

$$\begin{aligned} p &= 1.70951129135\dots, & c &= 0.16794779654\dots, \\ I &= 1.96959041447\dots, & D &= 1.56777020071\dots \end{aligned}$$

The second correction comes from a gamma-density transition near $y = \frac{1}{2} \log N$; it is not the full quadratic Taylor moment.

Proof. Put

$$P_N(s) = \frac{(1-s)^{N-1}}{1+s}, \quad h_N(t) = P_N(t/N) - 1 + t \quad (0 \leq t \leq N).$$

The elementary Taylor estimates

$$P_N(0) = 1, \quad P'_N(0) = -N, \quad 0 \leq P''_N(s) \leq N^2 - N + 2$$

give, for $N \geq 2$,

$$0 \leq h_N(t) \leq \min(t, t^2). \quad (6.3)$$

They also give the exact decomposition

$$A_N(b) = 1 + 2^{-b} - N3^{-b} - N4^{-b} + M_N(b), \quad (6.4)$$

where

$$M_N(b) = \mathbb{E}\{(1 + e^{-T_b})h_N(Ne^{-2T_b})\} \geq 0.$$

For $b = B\ell + d$, where d ranges over a fixed compact interval, center this last integral at $t_0 = \ell/2$. If $p_b(t) = t^{b-1}e^{-t}/\Gamma(b)$, then

$$\frac{M_N(b)}{p_b(t_0)} = \int_{-t_0}^{\infty} e^{-y}(1 + 2y/\ell)^{b-1}(1 + N^{-1/2}e^{-y})h_N(e^{-2y})dy.$$

For each fixed y the integrand converges to

$$e^{(2B-1)y}\{e^{-e^{-2y}} - 1 + e^{-2y}\}.$$

This passage is dominated, uniformly for bounded d . Indeed, choose $\varepsilon > 0$ so that $2 < 2B - 1 - \varepsilon < 2B - 1 + \varepsilon < 4$. The inequality $\log(1 + v) \leq v$ and (6.3) give an integrable majorant proportional to $e^{(2B-1-\varepsilon-2)y}$ on $y < 0$ and $e^{(2B-1+\varepsilon-4)y}$ on $y > 0$. The factor $1 + N^{-1/2}e^{-y}$ is at most two on the integration domain. Extending by zero below $-t_0$ completes the domination.

With $t = e^{-2y}$ the limiting integral equals

$$\frac{1}{2} \int_0^{\infty} (e^{-t} - 1 + t)t^{-\nu-1} dt = \frac{1}{2} \Gamma(-\nu). \quad (6.5)$$

Both integrations by parts used in the last equality are convergent because $1 < \nu < 2$; equivalently the value is $\Gamma(2 - \nu)/(2\nu(\nu - 1))$. Stirling's formula, uniformly for bounded d , gives

$$p_{B\ell+d}(\ell/2) = \sqrt{\frac{2B}{\pi}} (2B)^{-d} \frac{N^{-I}}{\sqrt{\ell}} \{1 + O(\ell^{-1})\}.$$

Consequently

$$M_N(B\ell + d) = D(d) \frac{N^{-I}}{\sqrt{\ell}} \{1 + o(1)\}, \quad D(d) = \sqrt{\frac{B}{2\pi}} \Gamma(-\nu) (2B)^{-d}. \quad (6.6)$$

Differentiating the gamma density inserts the score $\log(t_0 + y) - \psi(b)$, which tends to $-\log(2B)$. The same domination is valid after multiplication by this score. For $y \geq 0$ its absolute value is bounded by $C(1 + y)$. For $y < 0$, put $z = 1 + 2y/\ell \in (0, 1)$ and use $(-\log z)z \leq 1/e$ to replace z^{b-1} by z^{b-2} ; this only changes the exponent of the preceding majorant by $O(\ell^{-1})$. Hence, uniformly for bounded d ,

$$M'_N(B\ell + d) = -\log(2B)D(d) \frac{N^{-I}}{\sqrt{\ell}} \{1 + o(1)\}. \quad (6.7)$$

Let $f(d) = 2^{-d} - 3^{-d}$. Since $B \log 3 - 1 = p$, the first two nonconstant terms in (6.4) equal $N^{-p}f(d)$. The term $N4^{-b}$ and its derivative have order $N^{1-2p} = o(N^{-I}/\sqrt{\ell})$. The inequalities $p < I$ and $I < 2p - 1$ follow, for example, from

$$I - p = B \log(2B/3) - B + \frac{3}{2} > 0$$

(strict convexity of $x \log x - x + 1$), together with $12/5 < B < 5/2$, $\log 5 < 5/3$, and $\log 2 > 2/3$ for the second inequality. The latter crude bounds give $I < 13/6$ and $2p - 1 > 11/5$.

Now f' is positive below d_0 , negative above it, and

$$f(d_0) = c, \quad f''(d_0) = -(\log 2)Lq^{-p} < 0.$$

For every fixed $\delta > 0$, equations (6.4) and (6.7) give opposite derivative signs at $B\ell + d_0 \pm \delta$ for all sufficiently large N . The unique critical point therefore has $b_N = B\ell + d_0 + o(1)$. Substituting it into the stationary-point equation and using $f''(d_0) \neq 0$ yields (6.2). Its displacement changes the maximum value by $o(N^{-I}/\sqrt{\ell})$, so (6.4) and (6.6) give (6.1) for A_N . The exact eventual reduction Theorem 5.4 transfers it to C_N . $\square$

Corollary 6.4 (Concentration of almost maximizing orders). *Let $a_N, b_N^\circ > 0$ satisfy*

$$C_N - R_N(a_N, b_N^\circ) = o(N^{-p}).$$

Then

$$a_N = o(N^{1/2-p} \log N), \quad b_N^\circ = \frac{\log(Nq)}{\log(3/2)} + o(1).$$

Thus orders attaining the leading excess over one must approach the outer-order axis at a quantified rate.

Proof. The sharp asymptotic implies $R_N(a_N, b_N^\circ) > 1$ eventually, so the exclusion argument forces $a_N < 1$ and $b_N^\circ > 2 \log N$. The estimates in Lemma 5.3 in fact give

$$\partial_a R_N(a, b) \leq -\frac{1}{36\sqrt{N}(\log N + 4)}$$

on that region, because $\sqrt{N} > 432(\log N + 4)$ for $\log N \geq 24$. Integrating in a yields

$$C_N - R_N(a, b) \geq A_N - A_N(b) + \frac{a}{36\sqrt{N}(\log N + 4)}.$$

Both terms on the right are nonnegative, proving the bound on a_N . For each fixed $\varepsilon > 0$, the axis values at $B \log N + d_0 \pm \varepsilon$ have leading coefficients $f(d_0 \pm \varepsilon) < c$. Axis unimodality therefore excludes every b outside this interval from having $A_N - A_N(b) = o(N^{-p})$. Let $\varepsilon \downarrow 0$ to obtain the asserted location of b_N° . $\square$

Scope and next questions. The proof establishes an eventual theorem with a conservative explicit threshold; it does not establish $C_N = A_N$ at every $N \geq 2$. Determining the smallest such index, or proving equality at all indices, is a precise remaining question. A second question is an effective error bound in (6.1); dominated convergence gives the stated asymptotic but no small numerical error estimate at ordinary truncation indices. Expanding the centered gamma density suggests an inverse-logarithm series multiplying $N^{-I}/\sqrt{\log N}$, with coefficients given by derivatives of $\Gamma(-\nu)$, before still smaller powers of N . That longer expansion is not claimed here.

7 Polynomial separation and a larger uniform Lerch zero regime

The incoming effective-saturation report [3] obtains an explicit separation bound $(3n)^{-(n-2)}$ for the reciprocal-gamma Appell polynomials and a sufficient derivative order of size $n^{2n+O(n)}$. Two refinements give a polynomial sufficient order. First, a factorial diagonal multiplier converts a rational initial segment of the gamma product into a polynomial with quantitatively separated negative roots. Second, summing reciprocal distances from all roots replaces the coarser product estimate that charges every root the nearest distance. The resulting sufficient order is $O(n^6 \log^2 n)$, uniformly in the entire Lerch deformation interval. This is a new consequence of classical multiplier and mesh theory; the underlying preservation theorems are not new claims.

Write

$$\frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}, \quad W_\rho(t) = \frac{1}{1 - \rho e^{-t}}.$$

For $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, retain the manuscript's signed family

$$F_{n,k}^\rho(a) = (-1)^{n+k} D_{n,k}^\rho(a) = \int_0^\infty t^k e^{-at} W_\rho(t) Q_n(\log t) dt. \quad (7.1)$$

Thus $D_{n,k}^1 = \gamma_n^{(k)}$, and for $\rho < 1$ it is the k th a derivative of $\sum_{m \geq 0} \rho^m \log^n(a+m)/(a+m)$. Here the Stieltjes convention is $\zeta(s, a) = (s-1)^{-1} + \sum_{n \geq 0} (-1)^n \gamma_n(a) (s-1)^n / n!$. The established Appell–Laplace formula is [1, `zeros:thm:laplace`]; it also follows by differentiating $t^z/\Gamma(1+z)$ under the defining gamma integral. We recall the variation proof below so that the completeness of the zero list does not depend on a numerical root search.

7.1 A factorial multiplier and a rational separation bound

For a polynomial whose nonzero roots all have one sign, order their absolute values as $0 < r_1 \leq \dots \leq r_d$ and define its logarithmic mesh to be $\min_{i < d} r_{i+1}/r_i$. Ordinary mesh means the minimum additive distance between consecutive roots, counting multiplicity.

We use the following two classical preservation statements. A real-rootedness-preserving operator commuting with translations does not reduce ordinary mesh. A real-rootedness-preserving diagonal operator does not reduce logarithmic mesh on polynomials with all roots of one sign. The second statement is the theorem of Katkova, Shapiro and Vishnyakova [7]; the first includes the Riesz theorem for $1 + c\partial_x$. Both follow from the interlacing pencil criterion: apply the operator to every real linear combination of $p(x)$, $p(x+h)$, or $p(x)$, $p(qx)$, respectively. Interlacing of the image pair gives the asserted mesh inequality. Here h is below the original additive mesh and $1 < q$ below the original logarithmic mesh; passage to the supremum includes equality.

Lemma 7.1 (A concrete diagonal multiplier). *The operator*

$$\mathcal{M}\left(\sum_{r=0}^d a_r x^r\right) = \sum_{r=0}^d \frac{a_r}{(r+1)!} x^r$$

preserves real-rootedness. If p has strictly negative roots and positive leading coefficient, then $\mathcal{M}p$ has strictly negative roots and logarithmic mesh at least that of p .

Proof. Put $E = x\partial_x$ and, for integers $N \geq 1$, define

$$\mathcal{M}_N p(x) = \left[\prod_{j=2}^{N+1} (1 + E/j) \right] p(x/N).$$

Each factor preserves real-rootedness. Indeed, for positive integral j ,

$$(1 + E/j)p(x) = \frac{x^{1-j}}{j} \frac{d}{dx} (x^j p(x)).$$

Rolle's theorem makes the derivative real-rooted; the division merely removes $j - 1$ roots at zero. Positive dilation also preserves real-rootedness. On the monomial x^r the multiplier is

$$N^{-r} \prod_{j=2}^{N+1} (1 + r/j) = \frac{1}{(r+1)!} \prod_{h=2}^{r+1} (1 + h/N) \longrightarrow \frac{1}{(r+1)!}.$$

The empty product at $r = 0$ equals one. Thus $\mathcal{M}_N p$ converges coefficientwise to $\mathcal{M}p$. Its degree does not drop, and closure of real-rootedness proves the first assertion. The coefficients of a polynomial with negative roots and positive leading coefficient are strictly positive. Their images remain strictly positive, so the real roots of $\mathcal{M}p$ cannot be nonnegative. The classical diagonal mesh statement now applies. In particular, a logarithmic mesh strictly greater than one proves simplicity of the image. $\square$

Theorem 7.2 (Polynomial Appell separation). *For every $n \geq 3$, the polynomial Q_n has n distinct real roots and*

$$\text{mesh}(Q_n) \geq \Delta_n, \quad \Delta_n = \frac{4}{n(n-1)(n-2)}. \quad (7.2)$$

For $n = 2$, one may take $\Delta_2 = 2$.

Proof. Set

$$B_n(x) = \prod_{j=1}^{n-1} (1 + \partial_x/j) x^n, \quad P_n(x) = \prod_{j=1}^{n-1} (1 + jx).$$

Complementation of elementary symmetric functions gives the exact identity

$$B_n(x) = nxR_n(x), \quad R_n(x) = \mathcal{M}P_n(x). \quad (7.3)$$

For explicit verification, if e_r denotes the degree- r elementary symmetric function of $1, 2, \dots, n-1$, then

$$R_n(x) = \sum_{r=0}^{n-1} \frac{e_r}{(r+1)!} x^r, \quad e_{n-1-r}(1, 1/2, \dots, 1/(n-1)) = \frac{e_r}{(n-1)!}.$$

The negative roots of P_n have absolute values $1/(n-1), 1/(n-2), \dots, 1$. Its logarithmic mesh is $(n-1)/(n-2)$. By Lemma 7.1, the roots of R_n can be written as $-r_1, \dots, -r_{n-1}$ with

$$0 < r_1 < \dots < r_{n-1}, \quad \frac{r_{i+1}}{r_i} \geq \frac{n-1}{n-2}.$$

The coefficient ratio supplies an absolute scale:

$$\sum_{i=1}^{n-1} \frac{1}{r_i} = \frac{R'_n(0)}{R_n(0)} = \frac{1}{2} \sum_{j=1}^{n-1} j = \frac{n(n-1)}{4}.$$

Consequently $r_1 \geq 4/[n(n-1)]$. Every consecutive nonzero-root gap is at least $r_1/(n-2)$, and the remaining gap, from $-r_1$ to the single zero at the origin in (7.3), is r_1 itself. This proves $\text{mesh}(B_n) \geq \Delta_n$. For $n = 2$, $B_2 = x(x+2)$ directly gives mesh two.

For $N \geq n-1$ define the finite reciprocal-gamma Appell polynomial

$$Q_{n,N}(x) = \left[\prod_{j=1}^N (1 + \partial_x/j) \right] (x + \gamma - H_N)^n.$$

The first $n-1$ factors give a translation of B_n . Each remaining factor preserves real-rootedness and, by Riesz's theorem, does not decrease additive mesh. Hence $\text{mesh}(Q_{n,N}) \geq \Delta_n$. The Weierstrass product

$$\frac{1}{\Gamma(1+z)} = e^{\gamma z} \prod_{j \geq 1} (1 + z/j) e^{-z/j}$$

converges locally uniformly, so $Q_{n,N}$ converges coefficientwise to Q_n . Continuity of the ordered roots of monic real-rooted polynomials preserves the positive gap inequalities in the limit. This proves both the stated bound and simplicity. $\square$

The zero at the origin belongs only to the auxiliary B_n . No logarithmic mesh is assigned to that zero: the multiplicative argument is applied to R_n , and the final additive gap is handled separately. This distinction is essential to the proof.

7.2 The inherited global zero bound, with multiplicity

Lemma 7.3 (Laplace variation). *For $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, the function $F_{n,k}^\rho$ has at most n positive zeros counted with multiplicity.*

Proof. Here is the elementary argument from the manuscript [1, `zeros:lem:variation`]. If a nonzero real kernel g has m sign changes and its exponentially weighted polynomial moments exist, its Laplace transform G has at most m positive zeros. For $m = 0$ the transform has a strict constant sign. For $m > 0$, choose a sign-change point t_0 . The kernel $(t_0 - t)g(t)$ has $m - 1$ sign changes, and its transform is

$$(\partial_a + t_0)G(a) = e^{-t_0 a} \frac{d}{da} (e^{t_0 a} G(a)).$$

Induction bounds its zeros by $m - 1$ and precludes an identically zero transform. Any finite collection of zeros of G of total multiplicity M forces at least $M - 1$ zeros of the displayed transformed derivative, by Rolle's theorem with multiplicities. Thus $M \leq m$, completing induction, including nonvanishing.

The factor $t^k W_\rho(t)$ is strictly positive. By Theorem 7.2, $Q_n(\log t)$ has exactly n simple sign changes; the cases $n = 0, 1$ are immediate. Moreover $W_\rho(t) \leq W_1(t) \leq 1 + t^{-1}$, so $k \geq 1$ gives integrability at zero and the exponential controls infinity. These bounds justify all differentiations used in the induction. $\square$

7.3 Sign control using the whole root configuration

Let Y_k have gamma shape $k + 1$ and rate k . For a real number s that is not a root of Q_n , set

$$S_n(s) = \sum_{j=1}^n \frac{1}{|s - x_{n,j}|}, \quad m(s) = \lceil 1 + S_n(s) \rceil.$$

Here $x_{n,1} > \dots > x_{n,n}$ are the roots of Q_n .

Lemma 7.4 (Configuration-sensitive sign transfer). *If $k \geq 16m(s)^2$, then uniformly for $0 \leq \rho \leq 1$,*

$$\left| \frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} - 1 \right| \leq \sqrt{2/3} < 1. \quad (7.4)$$

Thus $F_{n,k}^\rho(ke^{-s})$ and $Q_n(s)$ have the same strict sign.

Proof. The exact gamma average obtained by setting $t = e^s y$ in (7.1) is

$$\frac{(ke^{-s})^{k+1} F_{n,k}^\rho(ke^{-s})}{k! W_\rho(e^s) Q_n(s)} = \mathbb{E} \left[\frac{W_\rho(e^s Y_k) Q_n(s + \log Y_k)}{W_\rho(e^s) Q_n(s)} \right].$$

For real u , factorization over the roots yields

$$\left| \frac{Q_n(s+u)}{Q_n(s)} - 1 \right| \leq \prod_j \left(1 + \frac{|u|}{|s - x_{n,j}|} \right) - 1 \leq e^{S_n(s)|u|} - 1.$$

Also

$$\left| \frac{d \log W_\rho(t)}{d \log t} \right| = \frac{t\rho}{e^t - \rho} \leq 1,$$

since $e^t - \rho \geq \rho t$. Therefore, writing $R = W_\rho(e^{s+u})/W_\rho(e^s)$, one has $R \leq e^{|u|}$ and $|R-1| \leq e^{|u|} - 1$. The absolute difference of the full integrand from one is at most $e^{(1+S_n(s))|u|} - 1 \leq e^{m(s)|u|} - 1$.

For completeness, the required gamma moment estimate is

$$\mathbb{E}(e^{m|\log Y_k|} - 1) \leq \{2(e^{4m^2/k} - 1)\}^{1/2} \leq \sqrt{2/3} \quad (k \geq 16m^2).$$

Indeed, the squared integrand is bounded by $(Y_k^m - 1)^2 + (Y_k^{-m} - 1)^2$ and Cauchy–Schwarz applies. The exact integer moments are

$$\mathbb{E} Y_k^r = \prod_{j=1}^r (1 + j/k), \quad \mathbb{E} Y_k^{-r} = \prod_{j=0}^{r-1} (1 - j/k)^{-1}.$$

Both moments with $r = m$ are at least one. For $r = 2m$, their logarithms are at most $3m^2/k$ and $4m^2/k$, respectively, using $\log(1+v) \leq v$ and $-\log(1-v) \leq 2v$ for $0 \leq v \leq 1/2$. The inverse moments exist because $2m \leq k$. This gives the first bound. Finally $4m^2/k \leq 1/4$ and $e^{1/4} \leq (1 - 1/4)^{-1} = 4/3$ give the second. Taking expectations proves (7.4) and also verifies absolute integrability of every expression used. $\square$

7.4 The polynomial threshold and the simultaneous zero limit

Theorem 7.5 (Uniform saturation at polynomial derivative order). *For $n \geq 3$ define the explicit integer*

$$\mathcal{K}_n = \left\lceil (16 + 6n(n-1)(n-2)H_{n-1})^2 \right\rceil. \quad (7.5)$$

For every integer $k \geq \mathcal{K}_n$ and every $0 \leq \rho \leq 1$, the function $D_{n,k}^\rho$ has exactly n positive zeros, all simple. Numbering them increasingly as $a_{n,1,k}^\rho, \dots, a_{n,n,k}^\rho$ and putting $c_{n,j} = e^{-x_{n,j}}$, one has

$$\left| \log \frac{a_{n,j,k}^\rho}{k c_{n,j}} \right| < \frac{8}{\sqrt{k}} \quad (1 \leq j \leq n). \quad (7.6)$$

In particular,

$$\mathcal{K}_n = 36n^6 \log^2 n (1 + o(1)), \quad \mathcal{K}_n \leq 36n^4(n-1)^2(n-2)^2 \leq 36n^8.$$

The exact two-zero theorem remains stronger at $n = 2$; for the stated location estimate alone, the same proof gives the sufficient value $k \geq 784$ with $\Delta_2 = 2$.

Proof. Let $\delta = \Delta_n$ and $\eta = 8/\sqrt{k}$. The threshold condition is precisely

$$\sqrt{k} \geq 16 + \frac{24H_{n-1}}{\delta}.$$

It implies $\eta < \delta/3$, because $H_{n-1} \geq 1$. The intervals $[x_{n,j} - \eta, x_{n,j} + \eta]$ are disjoint. At either endpoint $s = x_{n,j} \pm \eta$, the reciprocal-distance sum obeys

$$\begin{aligned} S_n(s) &\leq \frac{1}{\eta} + \frac{3}{2\delta}(H_{j-1} + H_{n-j}) \\ &\leq \frac{1}{\eta} + \frac{3H_{n-1}}{\delta}. \end{aligned}$$

For a root h positions away, this uses $|s - x_{n,j \pm h}| \geq h\delta - \eta \geq 2h\delta/3$. Consequently

$$m(s) \leq 2 + S_n(s) \leq 2 + \frac{\sqrt{k}}{8} + \frac{3H_{n-1}}{\delta} \leq \frac{\sqrt{k}}{4}.$$

Lemma 7.4 gives opposite endpoint signs for $F_{n,k}^\rho(ke^{-s})$ on each interval, uniformly in ρ . There is therefore a positive zero in each of the n corresponding disjoint a intervals. Lemma 7.3, which counts multiplicities, makes this list complete and every zero simple. The interval endpoints give (7.6).

The asymptotic size follows from $H_{n-1} = \log n + O(1)$. For the displayed elementary upper bound, use $H_{n-1} \leq n-1$ and $16 \leq 6n(n-1)(n-2)$ for $n \geq 3$. They give $16 + 6n(n-1)(n-2)H_{n-1} \leq 6n^2(n-1)(n-2)$; the square on the right is already integral, so the ceiling preserves the inequality. For $n = 2$, the threshold becomes $(16 + 24/2)^2 = 784$. $\square$

Remark 7.6 (Sharper finite thresholds from certified auxiliary roots). The proof uses only a lower bound δ on the Appell mesh. Thus every proved $\delta > 0$ gives the sufficient order $\lceil (16 + 24H_{n-1}/\delta)^2 \rceil$, with the same logarithmic location error $8/\sqrt{k}$. Exact rational isolation of all roots of B_n supplies such a δ by Riesz's theorem. The accompanying script verifies the polynomial identity, isolates all 135 auxiliary roots for $2 \leq n \leq 16$, and validates the additive and logarithmic mesh inequalities with rational brackets. The following finite thresholds use those certified gaps:

n	3	4	5	6	8	10	12	16
sufficient k	3629	12996	36442	85413	327915	926207	2152710	8065605.

They remain sufficient bounds, not least thresholds. No finite root calculation is needed for the all-index theorem or its growing-index corollary. The exact audit also checks 90 finite-multiplier coefficient identities and seven gamma-moment bounds.

Corollary 7.7 (A power-sized growing-index regime). *Fix $0 < c < 1$. For all sufficiently large integers k , simultaneously for*

$$3 \leq n \leq c \frac{k^{1/6}}{(\log k)^{1/3}}, \quad 1 \leq j \leq n, \quad 0 \leq \rho \leq 1,$$

the function $D_{n,k}^\rho$ has exactly n simple positive zeros and

$$\frac{a_{n,j,k}^\rho}{k e^{-x_{n,j}}} = 1 + O(k^{-1/2}).$$

The implied constant can be chosen independently of n, j, ρ .

Proof. The defining quantity in (7.5) is increasing in n . At the largest permitted index its ratio to k has limsup at most $c^6 < 1$: substitute $n \sim ck^{1/6}(\log k)^{-1/3}$ into $36n^6 \log^2 n$, noting $\log n \sim \frac{1}{6} \log k$. Hence $k \geq \mathcal{K}_n$ throughout the range for all sufficiently large k . Theorem 7.5 applies. Exponentiation of the uniform logarithmic bound gives the conclusion; for example, for $k \geq 64$ its relative error is less than $8e/\sqrt{k}$. $\square$

7.5 What has and has not been resolved

This argument strengthens the effective growing-index theorem from a logarithmic index range to a power-sized one. It gives a rigorous polynomial answer to the incoming report's question about substantially smaller explicit saturation orders. It does not determine the least uniform threshold, and the bound on the Appell mesh is not claimed optimal. In particular, the elementary endpoint still forces every uniform simplicity threshold to be at least $n - 1$. Indeed, at $\rho = 0$, writing $x = \log a$ gives

$$\frac{d^k}{da^k} \frac{\log^n a}{a} = a^{-k-1} \prod_{j=1}^k (\partial_x - j) x^n.$$

For $k \leq n - 2$, the right-hand polynomial is divisible by x^{n-k} , so its positive zero at $a = 1$ is multiple.

Further precise targets are: improve the n^{-3} Appell mesh bound; determine whether the least uniform simplicity threshold is $n - 1$; replace the absolute log-moment estimate by a signed concentration estimate to improve the $k^{-1/2}$ location error; and extend the configuration-sensitive argument to all-order offsets with a growing index. The lower-branch shape problem at index two belongs to a separate line of results already developed in the source reports. We do not count its previously established cases as new results here.

8 Source audit, reproducibility, and further research

8.1 What should change in the project status

Table 2 records the precise scope of the proved changes. The integration notes name the source paths and incoming archive members. They are proposals for integration; the source repository has not been modified.

The rational tetralogarithm identity should now be designated a theorem with an exact certificate. The two duplication formulas already in the manuscript make the displayed cross-base relation equivalent, so its status changes at the same time. This does not establish uniqueness among all rational tetralogarithm relations. Nor should the higher golden formulas be promoted merely because the weight-four certificate is successful.

Source target	Result established here	Scope that remains open
Rational tetralogarithm, <code>golden:eq:tetra</code>	Exact proof, Theorem 2.1; the equivalent cross-base form follows	Minimal certificate size and reduction to named classical specializations
Incoming <code>conj:euler-rate</code>	Matching asymptotic, second scale, and maximizing-order displacement, Theorem 6.3	Effective error constants and the full smaller-index optimization
Fractional continuation of integer Euler tails	Compensated exact kernel and both logarithmic sequences, Theorems 3.2 and 3.5	Joint endpoint regimes and optimal truncation of divergent series
Fixed-total late-error comparison	All- N domination for $a \geq 1$; a unique reversal for $0 < a < 1$, Theorems 4.1 and 4.4	Quantitative dependence of the crossing order on a, b
Effective Lerch saturation	$O(n^6 \log^2 n)$ sufficient order and a power-sized index range, Theorem 7.5	Least saturation order and optimal polynomial root separation

Table 2: Status changes relative to the pinned ProveIt source. Each row distinguishes a proved statement from a sharper question.

For the Euler problem, the incoming lower asymptotic was already proved [2]. The new work supplies the missing uniform upper argument, an exact eventual reduction, and more precise asymptotics. The large numerical threshold in the reduction is part of the theorem’s explicit scope. The unique maximum and its strictly increasing location on the axis hold at every $N \geq 2$; strict decrease of the global C_N is proved here only in the eventual range.

The new Lerch estimate improves a sufficient bound. It does not change the meaning of the least uniform simplicity threshold. The exact finite calculations concern auxiliary polynomials with rational coefficients, while the all-index theorem concerns reciprocal-gamma Appell polynomials with transcendental coefficients. The transfer between them is the proved mesh-preserving gamma-product limit, not a numerical comparison of their roots.

8.2 A correction and three important proof distinctions

The passage adjoining `vertical:eq:Im3` in `03-algebraic.tex` describes the imaginary part of $\text{Li}_3(1+i)$ as no longer elementary. The displayed representation through a hypergeometric constant does not prove such a nonreducibility assertion. A precise replacement is:

The transformations displayed here do not reduce the imaginary part to the elementary constants used here; the following formula expresses it through the central-binomial Gaussian tower constant.

The accompanying patch makes this limited wording correction and preserves the displayed identity. A proof of arithmetic nonreducibility would require a different theorem.

Three other distinctions matter for integrating the new material. First, the compensated density at $a + b < 1$ does not contradict the source’s finite signed-measure obstruction. The primitive and density are tested against functions that vanish at zero. A possible critical endpoint atom is therefore invisible, and the ordinary alternating boundary series is not being used in a region where it diverges.

Second, retaining only the total-order logarithmic sequence would omit the term

$$-\frac{b\zeta(b+1)\sqrt{\pi}}{2\Gamma(a-1)}N^{-1/2}L^{a-2}.$$

N	Axis maximizer b_N	$A_N - 1$	$N^p(A_N - 1)$
2	3.239719039892	$3.9230434864 \cdot 10^{-2}$	0.12830
10	6.746559248034	$3.2851480198 \cdot 10^{-3}$	0.16829
100	12.331892789531	$6.8858193177 \cdot 10^{-5}$	0.18071
10^4	23.792066269001	$2.5107487089 \cdot 10^{-8}$	0.17292
10^{12}	69.281926034579	$5.1420139643 \cdot 10^{-22}$	0.167984320879

Table 3: Numerical axis optimization using the positive centered integral. Full working values are supplied in `data/late_euler_optimizers.csv`. Only the last row lies beyond the explicit global axis-reduction threshold.

For $0 < b < 1$, it occurs between two successive integer inverse-log orders. The real-order extension must retain both sequences. When $b = m$ is integral and a is not, the combined coefficients grow as $\Gamma(j - a)$; the exact gamma index is significant for optimal-truncation work.

Third, formal relation spaces and numerical period spaces are different objects. The incoming work contains exact nonmembership statements for specified finite collections of shuffle or auxiliary relations [3]. Such statements can certify that one proposed derivation is insufficient. They do not exclude all functional identities, and they do not prove arithmetic independence of the evaluated constants. The Gaussian conjectures S_6, S_8, S_{10}, S_{12} remain unresolved by this article.

8.3 Exact checks and numerical illustrations

The exact replay for the tetralogarithm theorem reconstructs the rational functions from the forty-two rows, factors every $1 - f_i$, checks the factorizations by multiplication, and cancels all 230 touched tensor coordinates. It retains the rational prime 2. Discarding rational contents would change the vector space and invalidate the proof. The verifier also checks the endpoint polynomial, the original six arguments and coefficients, the one-parameter formula, and the $t = 1/3$ specialization.

At points where an auxiliary factor vanishes, the descent functional is defined by its values on independent generators and then extended linearly. It is not defined by evaluating a logarithm of that vanishing factor. Likewise, a term involving $\ell_1(1)$ is removed only after its coefficient has been proved zero. These two details are included in the proof and replay because overlooking either would produce an invalid symbolic calculation.

The fractional verifier checks all 64 integer-density specializations with $1 \leq a, b \leq 8$, along with eight coefficient groupings. Its interval engine encloses the half-order comparisons using only integer square roots and rational arithmetic. The dedicated threshold replay uses that same engine and records its dependency explicitly. It certifies the two signs needed to locate the unique transition.

The Lerch replay verifies the auxiliary polynomial identity at $2 \leq n \leq 16$ and isolates all 135 roots in exact rational brackets. It checks simplicity, additive and logarithmic gap inequalities, 90 finite-multiplier coefficient identities, and seven exact gamma-moment inequalities. These finite checks support normalization and give useful improved thresholds, but the all-index theorem follows from the analytic proof.

The Euler diagnostic program checks the derivative polynomials symbolically and evaluates the positive integral remainder $M_N(b)$ at high precision. Table 3 illustrates the axis optimization; its numbers are numerical approximations, not interval-certified maximizers.

The second asymptotic scale converges much more slowly than the leading normalized maximum. For example, at the leading maximizing location $B \log N + d_0$, the diagnostic $N^I \sqrt{\log N} M_N$ is still approximately 1.45355 at $N = 10^{10000}$, whereas its proved limit is $D = 1.56777020071 \dots$. The small exponent $4 - (2B - 1) = 5 - 2B > 0$ in the limiting positive

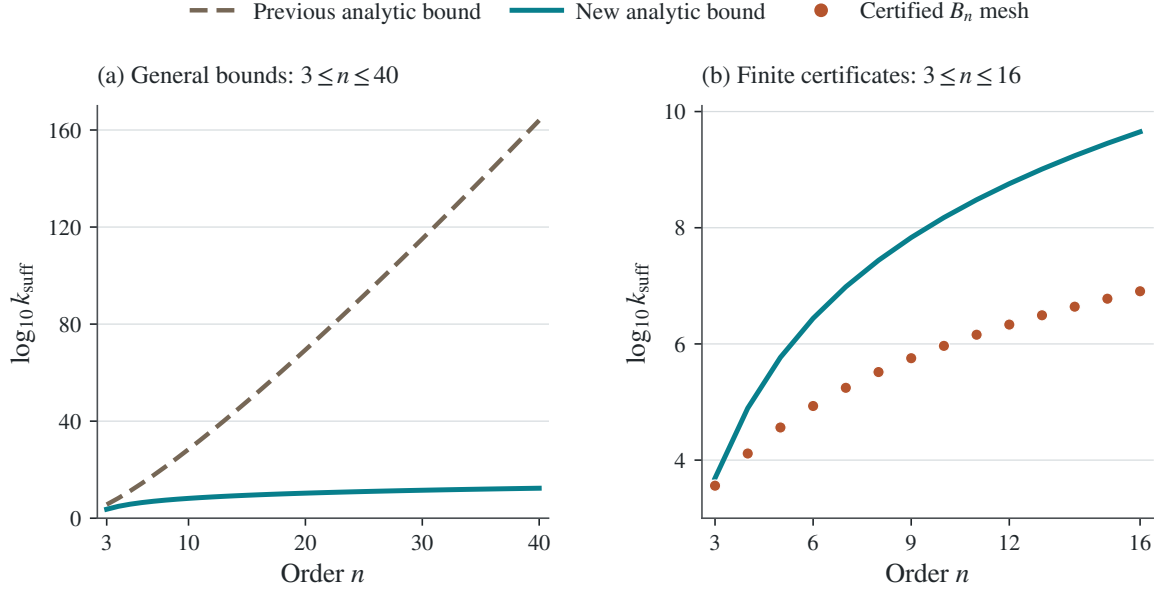


Figure 1: Uniform sufficient bounds for Lerch zero saturation. Panel (a) compares the earlier $576n^2(3n)^{2n-4}$ [3] with the new $\mathcal{K}_n$ from (7.5). Panel (b) compares $\mathcal{K}_n$ with the smaller bounds obtained from exact certified meshes of B_n . The vertical coordinates are base-ten logarithms. Every displayed quantity is a sufficient bound; no sharpness assertion is made.

tail explains why the dominated-convergence proof can coexist with this slow approach. The second term should therefore not be substituted into a rigorous finite- N error budget without an additional effective remainder estimate.

8.4 Further research questions and concrete conjectures

1. A shorter tetralogarithm derivation. Can the forty-two-row tensor certificate be decomposed into a small number of standard Kummer functional-equation substitutions? This would make the proof easier to recognize in the classical literature. The exact relation is established already; minimality is a separate optimization problem. A systematic library of real-branch descent certificates should record rational prime contents, argument factorizations, endpoint values, and permitted crossings.

2. Higher-weight golden and Gaussian identities. Apply the certificate method to the specific higher golden reconstructions, with their endpoint logarithms and higher-weight compatibility conditions made explicit. For the Gaussian problem, seek an exact certificate for S_6 before enlarging the conjectural S_{2m} family. A useful success criterion is an explicit rational relation in a stated functional-equation space whose analytic evaluation can be proved. A smaller numerical residual or formal rank calculation alone does not meet that criterion.

3. Is global axis reduction valid at every later index? The following strengthens the eventual theorem:

Conjecture 8.1 (Axis reduction at every $N \geq 2$). *For every integer $N \geq 2$,*

$$\sup_{a,b>0} R_N(a,b) = \max_{b>0} R_N(0,b).$$

Every positive outer order gives a strict inequality.

The axis maximum and its uniqueness are already proved for each such N . What is missing is a global comparison over the positive quadrant at the smaller indices. A proof need not show that $R_N(a, b)$ decreases in a for every a, b : that stronger pointwise statement is not used by the present argument. Validated compact optimization, joined to analytic estimates on all noncompact parameter regions, could also reduce the explicit threshold substantially.

4. Effective second-scale asymptotics and near maximizers. Obtain an explicit error estimate for (6.1), and develop the inverse-log N corrections multiplying $N^{-I}/\sqrt{\log N}$. The centered integral suggests coefficients involving derivatives of $\Gamma(1/2 - B)$. One must control the positive tail uniformly before integrating an expanded density. Determine also the sharp scale of the outer order among near maximizers; the proved bound in Corollary 6.4 is a necessary concentration estimate, not an optimal local expansion in a .

5. The reversal order near outer order one. For each fixed $b > 0$, the theorem implies $\nu(a, b) \rightarrow \infty$ as $a \uparrow 1$. Balancing the generic $N^{-1/2}$ difference and the critical N^{-1} difference leads to the following precise candidate.

Conjecture 8.2 (Near-critical reversal scale). *For fixed $b > 0$, with $\varepsilon = 1 - a \downarrow 0$,*

$$\nu(1 - \varepsilon, b) \sim \frac{(\log(1/\varepsilon))^{2b+2}}{\pi b^2 \Gamma(b+1)^2 \zeta(b+1)^2 \varepsilon^2}.$$

The heuristic balance is

$$\varepsilon b \zeta(b+1) \sqrt{\pi} \Gamma(b+1) \sqrt{N} \asymp L^{b+1}, \quad L = \frac{1}{2} \log N.$$

It uses $1/\Gamma(-\varepsilon) \sim -\varepsilon$ and $L \sim \log(1/\varepsilon)$ at the proposed crossing. This is a newly proposed asymptotic, not a consequence of exchanging limits in the fixed-parameter theorems. A proof needs a joint expansion that retains both terms uniformly as $a \uparrow 1$. Quantitative asymptotics for the derivative zero sequence ν_r from Corollary 4.5 form a related problem.

6. Optimal truncation and exponentially small Euler terms. For integral $b = m$ and nonintegral a , the coefficient law (3.16) identifies the factorial growth of the logarithmic series. Determine the optimal truncation index, a rigorous remainder near that index, and its relation to the exponentially small density terms discarded in the algebraic large- T expansion. Near integral a , the factor $\sin(\pi a)$ vanishes; any uniform result must resolve the transition from a divergent expansion to an exactly terminating one.

7. Better Appell mesh and the least Lerch threshold. Can the explicit n^{-3} gap be improved to cn^{-2} , or to a different optimal scale? The auxiliary factor $R_n(x) = \sum e_r x^r / (r+1)!$ provides a rational route to this question. A stronger uniform mesh estimate immediately improves the sufficient derivative order through $\lceil (16 + 24H_{n-1}/\delta_n)^2 \rceil$. The elementary endpoint gives the necessary condition $k \geq n-1$ for uniform simplicity; determine whether that condition is also sufficient. It is important to separate saturation in number from simplicity at the endpoint $\rho = 0$.

8. Signed concentration and uniform higher zero offsets. The present proof controls an absolute logarithmic moment and gives relative error $O(k^{-1/2})$. Cancellations in signed moments should permit sharper location estimates. Develop them while n grows with k and ρ ranges over the full closed interval. Fixed-index offset formulas in the manuscript are useful starting points, but their constants must be controlled with respect to n before a growing-index conclusion is drawn.

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